PRD: Ezra Ndinya Ogechi Portfolio Website

Product Requirements Document for the Antigravity build agent. Version 1.3 (final, hosted on Render), October 2026. Owner: Ezra Ndinya Ogechi.

1. Overview

Product: A personal portfolio website for Ezra Ndinya Ogechi, a Data Scientist and Machine Learning Engineer, that showcases his projects, sells his services, and promotes his ML course.

Brand and naming: The site's main name is **Ezra Ndinya Ogechi**, with the tagline *Data Scientist & ML Engineer*. The ML course, the free math lessons and all teaching content use the sub-brand **NeuralNest**, described as: "A welcoming home for learners exploring neural networks, AI, and machine learning." Example course title: "NeuralNest ML Course by Ezra Ndinya Ogechi". The footer copyright on every page reads "© 2026 Ezra Ndinya Ogechi. All rights reserved." The separate analytics site keeps the title "Ezra The Analyst". Ezra's TikTok handle (@codetointelligence4) is unchanged and appears only as a social link.

Goals

1. Present Ezra as a credible data scientist and ML engineer through real, verifiable work (GitHub projects, live demos, shipped apps).
2. Convert visitors into leads for three offers: (a) a full machine learning course, (b) ML web apps for clients, (c) native Android ML apps for clients.
3. Stay current with minimal effort. Ezra updates the site daily, and anyone with the link sees changes within minutes.
4. Let visitors choose between several visual themes.

Non-goals (v1)

- No invented statistics, fake testimonials, or placeholder reviews. Only show numbers and quotes that can be verified.
- No user accounts, payments, or course hosting on the site (the course section collects enquiries only).
- No separate admin dashboard.

Design inspiration: The look and feel should be comparable in polish to a modern gradient-based personal portfolio (rounded cards, strong hero, floating contact button, light/dark switching). The layout, section order, copywriting, and visual theme must be original and must not copy any existing site's text or structure.

2. Audience

Audience	Need
Prospective clients (businesses, startups)	Quickly see what Ezra can build and how to hire him
Prospective students	Understand the ML course, what they will learn, and how to enrol
Recruiters and hiring managers	Skim projects, tools, and GitHub activity
Fellow developers and learners	Browse repos and learn from the work

Primary device is mobile (the owner is in Kenya, and mobile-first traffic is expected), so the site must be fast on 4G.

3. Tech Stack and Architecture

- **Framework:** Next.js (App Router) with TypeScript, configured as a static export so it runs as a Render Static Site
- **Styling:** Tailwind CSS with CSS variables for theming
- **Theme switching:** `next-themes` (or equivalent) using a `data-theme` attribute on `<html>`
- **Hosting:** Render Static Site (free, always on, free custom domain), connected to the portfolio's GitHub repo. Every push to `main` triggers an automatic deploy.
- **Content source:** Markdown/JSON files committed to the portfolio repo (see Section 6)
- **Live GitHub data:** GitHub REST API. At build time, project dates and READMEs are fetched with a read-only personal access token stored as a Render environment variable (`GITHUB_TOKEN`). In the browser, the Currently Building strip calls the public GitHub API directly (no token) and caches the result for 10 minutes. Rebuilds run on every push to the portfolio repo, on every push to the project repos (a small GitHub Actions workflow calls the Render deploy hook), and every 6 hours as a safety net.

- **Contact form:** Formspree (free plan), posting straight from the page so no server is needed. It emails Ezra each enquiry. Spam protection: a honeypot field plus Formspree's built-in filtering.
- **Analytics:** Cloudflare Web Analytics, Plausible or Umami (privacy-friendly, no cookie banner needed)
- **Icons:** `lucide-react`
- **Animation:** Framer Motion, subtle only, respecting `prefers-reduced-motion`

Render hosting setup

The site is hosted on Render as a **Static Site**: free, always on, and with a free custom domain. A free Render Web Service is not used because it spins down after 15 minutes without traffic, so the first visitor after a quiet spell would wait for a slow restart. That would hurt a portfolio meant to impress clients.

Because a static site has no server, the build must follow these rules:

- Set `output: 'export'` in `next.config`. In Render, use Build Command `npm install && npm run build` and Publish Directory `out`, with auto-deploy from `main`.
- Set the environment variables `NODE_VERSION` and `GITHUB_TOKEN` in Render. Never prefix the token with `NEXT_PUBLIC_`.
- No API routes, server actions, middleware or ISR. Contact and course forms post directly to Formspree.
- The default `next/image` optimiser needs a server. Use `images: { unoptimized: true }` and pre-optimise images during the build (for example with `sharp`), or ship ready-made WebP files.
- Generate every page at build time: `generateStaticParams` for `/projects/[slug]`, static `sitemap.xml` and `robots.txt`, pre-generated social share images, and a custom `404.html`.
- Project pages and README text are fetched from GitHub during the build. The Currently Building strip loads in the visitor's browser.
- Every build uses Render pipeline minutes, so do not rebuild more often than needed.

Keeping the site fresh without a server

- Render rebuilds automatically on every push to the portfolio repo.

- Add this small GitHub Actions workflow to the `Machine_Learning_Projects` repo and each analytics repo, so a push there triggers a rebuild. Store the Render deploy hook URL as an encrypted secret named `RENDER_DEPLOY_HOOK`, never in the code.
- In the portfolio repo only, keep the `schedule` block as a safety net that rebuilds every 6 hours.

```
name: Refresh portfolio
on:
  push:
    branches: [main]
  schedule:
    - cron: "0 */6 * * *"
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - run: curl -fsS -X POST "${{ secrets.RENDER_DEPLOY_HOOK }}"
```

If Ezra later needs server features (for example private data or a custom API), the site can move to a paid Render Web Service without changing the content files.

Important repository note

All twelve current ML projects live as **sub-folders of one monorepo**: `https://github.com/Ezra-Data/Machine_Learning_Projects`. Therefore, repo-level GitHub topics cannot be used to categorise them. Categorisation is driven by the content manifest (Section 6). The GitHub API is used to enrich each project (last commit date for that folder path, README content, repo stats), not to discover them.

4. Information Architecture

Single-page home with anchored sections, plus dedicated pages for projects and the course.

Routes

- `/` Home
- `/projects` All projects with filters (Data Science / Data Analytics, category)
- `/projects/[slug]` Project detail page
- `/course` ML course page

- `/apps` Play Store apps (hidden from nav until the first app is published, see Section 5.8)
- `/contact` Contact page
- `/blog` (phase 2, optional)

Navigation: Home, About, Services, Projects, Course, Contact. On mobile: a hamburger menu that opens a full-screen sheet. The theme switcher sits in the header on every screen size.

Home page section order

1. Hero
2. Currently Building (live GitHub activity)
3. About
4. What I Offer (three offers)
5. Data Science Projects
6. Data Analytics Projects, then Featured Work (Chana spotlight)
7. Skills and Tools, then Certifications and Recognition
8. ML Course teaser, then Free Math Lessons teaser
9. Apps on Google Play (conditional)
10. Contact call to action
11. Footer

5. Functional Requirements

5.1 Hero

- Name: Ezra Ndinya Ogechi
- Role line: Data Scientist and Machine Learning Engineer
- A short original positioning statement (see copy guidance in Section 8)
- Two buttons: **See my work** (scrolls to projects) and **Work with me** (goes to contact)
- A small secondary link: **Join the ML course**
- Ezra's portrait and the blue data-orb visual, placed as specified in Section 16
- A subtle animated background that matches the active theme (gradient mesh or soft particles). It must be disabled when `prefers-reduced-motion` is set.

5.2 Currently Building (live strip)

- Shows the 3 to 5 most recent commits or updated project folders from the GitHub monorepo, with relative time (for example, "updated 3 hours ago").
- Fetched in the visitor's browser from the public GitHub API and cached for 10 minutes, with the build-time snapshot as a fallback.
- Each item links to the GitHub commit or folder.
- Graceful fallback: if the API fails or rate-limits, hide the strip silently rather than showing an error.

5.3 About

- 2 to 3 short paragraphs in Ezra's voice, written fresh and not reusing phrasing from any reference site.
- Must cover: his background in data analysis (Excel, Power BI, SQL), his move into machine learning engineering, that he builds ML systems and apps destined for Google Play, and that he teaches.
- A compact facts row: location (Kenya), focus areas, languages and tools. Only include facts Ezra confirms.

5.4 What I Offer

Three cards, each with a short description, three to four bullet points, and a call to action:

1. **Learn Machine Learning:** a full ML course (links to `/course`)
2. **ML Web Apps:** custom web applications that put models in front of real users (links to contact with `?service=web` prefilled)
3. **Native ML Apps:** Android apps with on-device or API-backed ML, published to Google Play (links to contact with `?service=native` prefilled)

Optional fourth line, shown as a smaller strip: Data analysis and dashboards (Excel, Power BI, SQL).

5.5 Projects: Data Science and Data Analytics

Two clearly separated sections on the home page (3 featured cards each, with a **View all** link) and a full `/projects` page.

Project card contents

- Title

- Category tag (Classification, Regression, Recommendation, Clustering, NLP)
- Short summary (1 to 2 sentences)
- Algorithm and tool chips
- Key result (a metric) if Ezra provides one, otherwise omit the line
- Buttons: **View details** and **GitHub** (and **Live demo** when a link exists)
- Last updated date, taken from GitHub

Filtering on `/projects`: top-level toggle between Data Science and Data Analytics, then category chips. Filtering is client-side and instant. The selected filter is reflected in the URL (`?type=ds&cat=nlp`) so views are shareable.

Project detail page (`/projects/[slug]`) must include: problem statement, dataset, approach, models compared, results, key learnings, tools, links (GitHub folder, demo), and the project's GitHub README rendered below as "Technical README" (fetched via the API, sanitised markdown rendering). Include previous and next project navigation.

Cover images: Use a generated, theme-aware cover per category (an abstract gradient with a category icon) until Ezra supplies real screenshots. Do not use a single letter on a gradient as the cover.

5.5b Data Analytics section and the linked analytics site

The Data Analytics section is populated at launch from Ezra's existing analytics site, https://ezra-wl21.onrender.com/ ("Ezra The Analyst"). That site stays live as a separate site, and the portfolio links to it prominently.

- **Section header card:** title, one-line description (Excel, SQL, Power BI, Python), and a large button **Visit my Data Analytics site** that opens in a new tab with `rel="noopener"`. Repeat the button at the top of the Data Analytics tab on `/projects` and in the footer.
- **Project cards:** rebuild the seven projects below natively in the portfolio, using the same card style as Data Science and a tool tag (SQL, Power BI, Excel, Python) for filtering. Do not embed the analytics site in an iframe.
- **Links:** each card's main button opens the project's GitHub page. A secondary link, **See on analytics site**, opens the analytics site.
- **Images:** copy the project images from the analytics site into `public/images/analytics/` instead of hotlinking them, so the portfolio does not break if that site goes down.

- **Content files:** analytics projects use `track: data-analytics` and a full `githubUrl` field, because they live in separate repos rather than in the ML monorepo.

#	Project	Tool	What it covers (from the analytics site)	GitHub
1	Data Cleaning in SQL	SQL (MySQL)	Cleaning a layoffs dataset into a structured, analysis-ready form	SQL_DATA_CLEANING_PROJECT_1
2	Data Exploration in SQL	SQL	Exploratory analysis to find trends, patterns and relationships	Exploratory_Data_Analysis.sql
3	Salary Survey Dashboard	Power BI	Interactive dashboard on a survey of 630 data professionals: countries, languages, salaries by title, work-life balance	PowerBI_Salary_Survey_Analysis_Dashboard
4	Excel Data Project	Excel	Data cleaning, pivot analysis and a dashboard on demographic and economic effects on buying behaviour	Excel_Data_Project
5	Pandas EDA: Global Population	Python (Pandas)	Exploration of population trends across continents over time	EDA_IN_PANDAS.ipynb
6	Movie Correlation Analysis	Python (Pandas)	How budget and votes relate to a movie's gross revenue	Movie Correlation Project
7	Sales Tracking Dashboard	Power BI	Monthly shipments, total revenue, customer listing and a geographic sales map	Sales_Dashboard_in_powerbi

Analytics site fixes (made by Ezra on 9 October 2026; the build should verify them on the live site):

- The footer copyright now names Ezra Ndinya Ogechi instead of "Untitled". Check every page of that site (index, elements, generic), because the footer is repeated on each one.
- The Power BI Salary Survey card should link to its GitHub repo, not a local file path. The portfolio always uses the GitHub link.
- The analytics site shows contact details that differ from the portfolio's. The portfolio uses only the details in Section 5.9.

5.5c Featured Project: Chana

Chana is a campus chatbot and navigation system for Dedan Kimathi University of Technology (DeKUT). Ezra's team built it at a hackathon and placed **First Runner-Up**. It is the portfolio's headline project, so it gets a spotlight on the home page right after the Data Science and Data Analytics project sections.

- **Home spotlight:** a wide card placed directly after the Data Science and Data Analytics project sections. It shows the name Chana, the line "Campus chatbot and navigation system for Dedan Kimathi University of Technology", a **First Runner-Up** badge with the event name and year, tool chips from the content file, and buttons for **View project**, **GitHub** (`https://github.com/Ezra-Data/Chana`) and **Live demo** (only when a demo link exists).
- **Detail page** `/projects/chana` : the standard project layout (problem, approach, result, tools, links) plus a **Recognition** block (event, organiser, placement, date, Ezra's role, team members), screenshots or a demo video, and the GitHub README rendered below it when the repository is public.
- **Content file** `content/projects/chana.md` , with front matter fields: `title: Chana` , `slug: chana` , `track: data-science` , `category: AI Applications` , `featured: true` , `spotlight: true` , `githubUrl: https://github.com/Ezra-Data/Chana` , `award: First Runner-Up` , `event` , `eventDate` , `role` , `team` , `demoUrl` , `proofUrl` .
- **Listing:** Chana also appears on `/projects` under Data Science, with a new category chip, **AI Applications**.
- **No invented details:** until Ezra supplies the write-up, show only the facts above. Do not invent features, technologies, accuracy figures, user numbers or outcomes.
- **Proof of placement:** when Ezra supplies a certificate, photo or announcement link (`proofUrl`), show it as a **View proof** link in the Recognition block.

- **Repository access:** when checked on 10 October 2026, `https://github.com/Ezra-Data/Chana` returned "not found", which usually means the repository is private or the address differs. The build must always show the GitHub button when `githubUrl` is set, so the link never silently disappears, and hide only the README block if the GitHub API returns 404 or 403. While the repository is private, also show **Source code available on request** with a link to the contact form instead. Ezra should make the repository public (after checking it contains no passwords or API keys, including in past commits) or confirm the correct address.

5.5d Certifications and Recognition

- **Home:** a compact strip of up to three cards, newest first, with a **View all** link to `/certifications`. Add **Certifications** to the footer quick links.
- **Page** `/certifications`: two groups. **Awards** (Chana, First Runner-Up) and **Certificates** (the GOMYCODE hackathon certificate and the St. Ann's College transcript). Hide a group, or the whole section, when it has no entries.
- **Card contents:** title, issuer, date, a type label, a one-line description, a preview thumbnail, and buttons **View certificate** (opens the PDF in a new tab) and **Verify** (opens the issuer's record page).
- **Truthful labels:** show each item's type exactly as issued. The GOMYCODE document is a **Certificate of Participation** and states that it is not a prize or professional qualification, so the site must never call it an award, a qualification, or say Ezra is "certified".
- **Content file** `content/certifications.json`, one object per entry: `title`, `issuer`, `date`, `type` (`award`, `participation` or `course`), `description`, `file`, `thumbnail`, `verifyUrl`, `projectSlug`, `team`, `projectName`.

#	Entry	Type	Details
1	Chana, First Runner-Up	Award	Hackathon name, organiser and date to be supplied by Ezra. Links to `/projects/chana`.
2	Come Build with AI hackathon (GOMYCODE, first edition)	Participation	Issued by GOMYCODE for the hackathon on 27 September 2026, Kenya / Online. Presented to Ezra Ndinya Ogechi as a listed member of Team WAHENGA, project BIA.

#	Entry	Type	Details
3	Internet and Computing Core Certification, academic transcript (St. Ann's College)	Course	Department of Information Technology, completed 6 June 2023. Three modules, each graded Credit: Computer Fundamentals (73), Office Applications (73) and Living Online (77).
4	School certificate (Kisii)	Education	Ezra has added the file to the certificates folder. Show only what is printed on the document (title, school, year, grades). Do not add or assume anything else, and do not display index or registration numbers as text.

GOMYCODE certificate files

- PDF: `public/certificates/gomycode-come-build-with-ai-2026.pdf`
- Preview image: `public/certificates/gomycode-come-build-with-ai-2026.png`. A static site cannot render a PDF thumbnail, so Ezra supplies a PNG export of the certificate.
- Alt text: "Certificate of Participation awarded to Ezra Ndinya Ogechi by GOMYCODE for the Come Build with AI hackathon, 27 September 2026."
- **Verify** link: use the exact record address printed on or behind the certificate's QR code (the certificate points to hackathon.gomycode.com). Ezra supplies the full address, and the build must not guess it. Until it is supplied, hide the Verify button.

St. Ann's College transcript

- Files: `public/certificates/st-anns-college-icc-transcript-2023.jpg` for the preview. A PDF or clean scan is preferred over a phone photo.
- Show it as an **Academic transcript** with the three modules and their grades. The document carries Certiport branding, but it is the college's transcript, so do not describe it as a Certiport-issued certificate unless Ezra also supplies that separate certificate.
- Privacy: do not publish the registration number as text, and crop or blur it in the preview image. The page shows the name, programme, college, completion date and module results only.

- Alt text: "Academic transcript for Ezra Ndinya Ogechi from St. Ann's College, Internet and Computing Core Certification, completed 6 June 2023."

Wording on the Certifications page

- Do not use "Verified credentials" as the small heading above the page title. Use "Recognition and Certificates", because participation certificates and an award without proof are not verified credentials.
- Replace the intro sentence about all entries being "genuine, confirmed participation" with a plain one, for example: "Hackathon results, certificates and training I have completed."
- Use the word "verified" only on an entry that has a working `verifyUrl`, and then only next to the Verify button.

5.6 Skills and Tools

Grouped, not a flat list:

- **Data:** Excel, Power BI, SQL, Pandas, NumPy
- **Machine Learning:** Scikit-learn, XGBoost, TensorFlow (confirm), model evaluation, feature engineering
- **Deployment and Apps:** Streamlit/Flask/FastAPI (confirm), TensorFlow Lite (confirm), Android (confirm framework: Kotlin, Flutter, or React Native)
- **Other:** Git and GitHub

Ezra will confirm the final list. Do not display skill-level percentages or progress bars.

5.7 ML Course

A teaser on the home page and a full `/course` page containing:

- Who the course is for and prerequisites
- Syllabus by module (draft below, Ezra to finalise)
- Format (online or in person, live or recorded), duration, schedule, and price, or "enquire for pricing"
- What students build (can reference the real projects in this portfolio)
- Instructor section
- An **Enquire / Join waitlist** form (name, email, WhatsApp number, experience level) that sends an email to Ezra

Draft syllabus (editable by Ezra)

1. Python and data foundations (Pandas, NumPy)
2. Data cleaning and exploratory analysis
3. Regression models (linear, ridge, lasso, tree-based)
4. Classification models (logistic regression, SVM, decision trees, random forests, XGBoost)
5. Clustering and recommendation systems
6. Natural language processing basics
7. Model evaluation, tuning, and avoiding overfitting
8. Deploying a model as a web app
9. Putting a model into an Android app
10. Capstone project

5.8 Apps on Google Play

- Section appears only when at least one app entry exists in `content/apps.json`.
- Each app card shows the icon, name, one-line description, screenshots carousel, and the official Google Play badge linking to the store listing.
- Until the first app is published, the section and nav link are hidden. A small line in About can say apps are in development.

5.9 Contact

- Form fields: name, email, service interest (dropdown: ML Course, Web App, Native App, Data Analysis, Other), message.
- Direct options: email, WhatsApp click-to-chat link, LinkedIn, GitHub. (Ezra to supply contact details he wants public.)
- A **floating contact button** on all pages that opens a small panel offering Message, WhatsApp, and Email. It must not cover important content on small screens, and it must be keyboard accessible.
- Success and error states for the form, with client-side validation and Formspree's spam filtering.

Public contact details and links (confirmed by Ezra)

- WhatsApp: +254 738 933 046, with the click-to-chat link `https://wa.me/254738933046`
- LinkedIn: https://www.linkedin.com/in/ezra-ogechi-1a615534a/

- GitHub: https://github.com/Ezra-Data
- X: https://x.com/OgechiEzra
- Reddit: https://www.reddit.com/user/Ogechi_Ezra7054/
- TikTok: https://www.tiktok.com/@codetointelligence4
- Analytics site: https://ezra-wl21.onrender.com/
- Email: ogechiezranalyst@gmail.com, as a mailto link with the subject prefilled by service. Show it on the contact page, in the floating contact panel, in the footer icons and in the Contact call to action. Formspree form notifications are sent to this address.

Show the social icons in this order in the footer and on the contact page: GitHub, LinkedIn, X, TikTok, Reddit, WhatsApp, Email. All external links open in a new tab with `rel="noopener noreferrer"`.

5.10 Footer

Name, short tagline, quick links, social icons (GitHub, LinkedIn, email), and copyright. No duplicated link groups.

6. Content Management (daily updates)

Ezra updates the site by editing files and pushing to GitHub. No code changes required for routine updates.

Content folder structure

```
content/
  site.json            # name, tagline, socials, contact, hero text
  about.md
  skills.json
  course.json          # syllabus, price, format, status
  apps.json            # Play Store apps (empty at launch)
  certifications.json  # awards and certificates (Section 5.5d)
  testimonials.json    # real quotes only (hidden while empty)
  projects/
    credit-card-fraud-detection.md
    ... one file per project
```

Project file format (front matter plus body)

```
---
title: Credit Card Fraud Detection
slug: credit-card-fraud-detection
track: data-science        # or data-analytics
category: Classification   # Classification | Regression | Recommendation |
Clustering | NLP | (analytics categories)
summary: One or two sentence summary.
tools: [Python, Scikit-learn, Pandas]
algorithms: [Logistic Regression]
githubPath: Credit_Card_Fraud_Detection_Logistic_Regression
demoUrl:
result:
featured: true
order: 1
published: true
---
Problem, dataset, approach, results, learnings (markdown body).
```

Rules

- `githubPath` is the sub-folder inside `Ezra-Data/Machine_Learning_Projects`. The build must construct the folder URL and API calls from it. For future projects in other repos, support an optional `repo` field.
- Setting `published: false` hides a project.
- A push to `main` redeploys automatically on Render (about 1 to 3 minutes). Project dates and READMEs refresh on the next rebuild (triggered by pushes, plus every 6 hours). The Currently Building strip refreshes within 10 minutes.
- Provide a short `README.md` in the portfolio repo explaining how to add a project, update the course, and add an app.

7. Project Catalogue (launch content)

All twelve ML projects are in the **Data Science** track. The Data Analytics track holds seven projects that already live on Ezra's analytics site (Section 5.5b), so both tracks are populated at launch.

Base repository: `https://github.com/Ezra-Data/Machine_Learning_Projects/tree/main/`

#	Title	Category	Folder (`githubPath`)	Algorithms (from folder name; Ezra to confirm)
1	Credit Card Fraud Detection	Classification	Credit_Card_Fraud_Detection_Logistic_Regression	Logistic Regression
2	Customer Churn Prediction	Classification	Customer_Churn_Prediction_System_Dtree_Rfc_XGBC	Decision Tree, Random Forest, XGBoost
3	Breast Cancer Classification	Classification	Breast_Cancer_Classification_Prediction_Logistic_Regression	Logistic Regression
4	Parkinson's Disease Detection	Classification	Parkinson_s_Disease_Detection_SVM	Support Vector Machine
5	House Price Prediction	Regression	House_price_prediction_Using_LR_Ri_L_Ra_HGBR	Linear, Ridge, Lasso, Random Forest (confirm), HistGradientBoosting
6	Car Price Prediction	Regression	Car_Price_Prediction_Using_Linear_Regression	Linear Regression
7	Medical Insurance Cost Prediction	Regression	Medical_Insurance_Prediction_LinearRegression	Linear Regression
8	Gold Price Prediction	Regression	Gold_Price_Prediction_RandomForestRegressor	Random Forest Regressor
9	Movie Recommendation System	Recommendation	Movie_recommendation_system	Confirm (likely content-based)
10	Customer Segmentation	Clustering	Customer_Segmentation_Kmeans	K-Means
11	Fake News Classification	NLP	Fake_News_Prediction_Using_Logistic_Regression	Logistic Regression, text features
12	Spam Message Classification	NLP	Spam_Mail_Predictions_Logistic_RegressionClassification	Logistic Regression, text features

Handling missing descriptions: Until Ezra supplies write-ups, the build agent must generate a neutral one-line summary from the title and algorithms only, and must **not** invent datasets, accuracy figures, or business outcomes. Fields such as `result` and `demoUrl` stay empty and their UI lines are hidden.

Featured on the home page at launch (suggested): Customer Churn Prediction, Credit Card Fraud Detection, Movie Recommendation System. Ezra can change this by editing `featured`.

8. Copywriting Guidance

- First person, warm, confident, and plain. No buzzword stacking.

- Do not reuse any wording or structure from the reference site (for example its hero line, "Delivering Results That Matter", "Want a Similar Project?", or its stat blocks).
- Say concretely what Ezra does: turns messy data into models, builds the apps that put those models in people's hands, and teaches others to do the same.
- Mention the Play Store ambition honestly ("building ML apps for Android") until real apps are live.
- Provide draft copy for every section in the build, marked clearly so Ezra can edit it in `content/`.
- Tone examples for the hero: a short statement about data to model to app, not a list of keywords.

9. Design Requirements

9.1 Visual direction

Modern, spacious, card-based, with rounded corners (16 to 24px), soft shadows or glows, a clear type hierarchy, and generous whitespace. Visually comparable in quality to a modern gradient portfolio, but with a different identity (palette, hero, and layout).

9.2 Theme system

Four themes, switchable at any time, implemented as CSS-variable token sets under `[data-theme="..."]`.

Theme	Character	Background	Accent
Midnight Aurora (default dark)	Deep navy with aurora glow	#0B1020	Teal #2DD4BF and violet #8B5CF6
Daylight (default light)	Clean, bright, minimal	#F8FAFC	Indigo #4F46E5 and sky #0EA5E9
Ember	Warm charcoal	#16120F	Amber #F59E0B and coral #FB7185
Terminal	Black, code-inspired, monospace headings	#050805	Green #22C55E

Behaviour

- A theme switcher in the header (an icon button opening a small menu, with a colour swatch for each theme).
- First visit: follow the device preference (`prefers-color-scheme`): Midnight Aurora for dark, Daylight for light.
- The choice persists per visitor (localStorage), applied before first paint to avoid a flash of the wrong theme.
- Define semantic tokens only (`--bg`, `--surface`, `--text`, `--muted`, `--accent`, `--accent-2`, `--border`, `--ring`) so components never hard-code colours.
- All themes must meet WCAG AA contrast for text.
- Smooth, short transition between themes (150 to 250 ms), disabled for reduced-motion users.

9.3 Typography

A modern sans-serif (Inter or Plus Jakarta Sans) for body and headings. The Terminal theme uses a monospace font (JetBrains Mono) for headings. Load fonts with `next/font` for performance.

9.4 Responsive behaviour

Mobile-first. Breakpoints at 640, 768, 1024, and 1280 px. Cards stack on mobile, 2 columns on tablet, 3 on desktop. Tap targets at least 44 px.

10. Non-Functional Requirements

- **Performance:** Lighthouse Performance 90 or higher on mobile. LCP under 2.5 s on a 4G connection. Use `next/image`, lazy loading, and minimal client JavaScript.
- **Accessibility:** WCAG 2.1 AA. Semantic HTML, keyboard navigation, visible focus rings, alt text, correct heading order, and a skip-to-content link.
- **SEO:** Per-page titles and descriptions, Open Graph and Twitter cards (a generated OG image per project), `sitemap.xml`, `robots.txt`, and JSON-LD `Person` schema.
- **Security:** No secrets in client code. The GitHub token is read-only, used only at build time, and never sent to the browser. Sanitise rendered README markdown. Rely on a honeypot field and Formspree's spam filtering for forms.
- **Reliability:** If the GitHub API is unavailable, the site still renders fully from local content, because GitHub data is an enhancement.
- **Privacy:** Cookie-free analytics. A short privacy note in the footer for the contact form.

11. Milestones

1. **Foundation:** Next.js project, Tailwind, theme tokens, four themes with switcher, layout, header, footer, and floating contact button.
2. **Content system:** Markdown/JSON content loading, project manifest for all 12 projects, and the project card and filter components.
3. **GitHub integration:** Currently Building strip, last-updated dates, and README rendering, with build-time data, a live in-browser strip, and fallbacks.
4. **Pages:** Home, About, Services, Projects (both tracks), Project detail, Course, and Contact (with working form).
5. **Polish:** SEO, accessibility audit, performance tuning, and OG images.
6. **Deploy:** Render Static Site connected to GitHub, environment variables set, deploy-hook workflows added, custom domain (if available), and the update guide README.

12. Acceptance Criteria

1. All 12 projects appear under Data Science, correctly categorised and filterable, with working GitHub links to the right sub-folders.
2. The Data Analytics section shows the seven analytics projects and a link to the analytics site (Section 5.5b), and new analytics projects added to the content folder appear automatically.
3. Adding a new project markdown file and pushing to `main` makes it visible on the live site within about 2 minutes, with no code changes.
4. The Currently Building strip updates within 10 minutes of a new push to the projects repo.
5. A visitor can switch between all four themes; the choice persists across reloads and there is no theme flash on load.
6. The site contains no fabricated stats, testimonials, or results.
7. Contact and course enquiry forms deliver email to Ezra and show clear success and error states.
8. The Apps section stays hidden until `apps.json` has an entry.
9. Lighthouse mobile scores: Performance 90 or higher, Accessibility 95 or higher, SEO 95 or higher.
10. No text, section titles, or layout from the reference site are reused.

13. Inputs Still Needed from Ezra

- Transparent-background or vector (SVG) versions of the two logo images, if Ezra has them (Section 16)
- Nothing further for contact details: email, WhatsApp and all social links are confirmed (Section 5.9)
- Project write-ups (problem, dataset, approach, result, demo link) for each of the 12 projects
- Short outcome write-ups for the seven analytics projects (optional, can be added later)
- Confirmed algorithms for Movie Recommendation, House Price, and any others marked "confirm"
- ML course details: final syllabus, format, duration, price or enquiry method, start dates
- Android approach (Kotlin, Flutter, or React Native) and any apps already on Google Play
- Domain name, if he has one (otherwise start on a free `.onrender.com` address)
- A free Formspree account for the contact and course forms
- A GitHub personal access token (read-only, public repo scope) for the Render build environment

14. Risks and Mitigations

Risk	Mitigation
GitHub API rate limits	Authenticated build-time requests, 10-minute browser caching, and graceful fallback to local content
Monorepo structure limits topic-based discovery	Manifest-driven projects with `githubPath` (Section 6)
Thin project descriptions at launch	Neutral generated summaries, with fields hidden when empty, and no invented claims
Theme flash on load	Inline script or `next-themes` setting the attribute before first paint

Risk	Mitigation
Contact form spam	Honeypot field and Formspree's spam filtering

15. Visual Impact and Customer Conversion

The site must look striking and also turn visitors into enquiries. Everything below applies to all four themes.

15.1 Visual impact

- **Hero:** two columns on desktop (headline and buttons on the left, portrait in a rounded frame with a soft glowing ring over the data-orb visual on the right). Stacked on mobile.
- **Look:** aurora gradient background, glass-style cards with a blurred backdrop, gradient accent words in headings, card lift on hover, and fade-up on scroll.
- **Motion with purpose:** a rotating role line under the name (Data Scientist, ML Engineer, Educator, App Builder), count-up numbers, and a soft cursor glow on desktop. All motion switches off for visitors who prefer reduced motion.
- **Themes with personality:** each theme changes glow, hero treatment and accent, not only colours. Terminal gets a typed-code intro line, Ember gets warm glow, and so on.
- **Budget:** animation code under about 30 KB, no layout shift, and the hero image loads with `priority`.

15.2 Trust without invented numbers

- Show only counters computed from real data: number of ML projects, number of analytics projects, and commits in the last 30 days from GitHub. These update themselves, so they cannot go stale or be exaggerated.
- Add `content/testimonials.json`. The testimonials section stays hidden until it has at least one real entry, then appears automatically.
- Show tool badges (Python, SQL, Power BI, Excel, Scikit-learn, and so on) in the hero area as a quick credibility row.

15.3 Make it easy to hire

- A **Work with me** button stays in the header on every page. On mobile, a sticky bottom bar offers **WhatsApp** and **Message**, hidden on the contact page and while the keyboard is open.
- WhatsApp links prefill the message by service, for example "Hi Ezra, I'm interested in a native ML app."
- Contact form: name, email, service, message. It states the expected reply time (Ezra sets this) and sends an automatic confirmation email to the visitor.
- Optional **Book a 15-minute call** button if Ezra supplies a Cal.com or Calendly link.
- A **How it works** strip: Discuss, Scope, Build, Launch and support.
- An FAQ accordion covering timelines, how pricing works, tools used, code ownership, support after launch, and course format. Ezra provides the answers; the build may draft placeholders clearly marked for review.
- Use **Get a quote** wording instead of showing prices unless Ezra supplies packages.

15.4 Course conversion

- Course page order: outcome headline, who it is for, what students build (linking to real projects here), syllabus accordion, format and dates, instructor, enquiry form, WhatsApp button.
- Phase 2 ideas: a free preview lesson or an ML project checklist as a lead magnet, and mobile-money payment for enrolment.

15.5 Projects as case studies

- Every project page ends with a call to action, **Need something like this? Talk to me**, with the matching service prefilled in the contact form.
- Embed or link live demos (for example Streamlit or Render apps) when Ezra provides them, lazy loaded.

15.6 Measurement

Track these events in the analytics tool: CTA clicks, WhatsApp clicks, contact form submissions, course enquiries, analytics-site link clicks, and theme choice. Ezra can then see which sections bring in leads.

16. Brand Assets

Ezra is adding these files to the portfolio folder. Keep untouched originals in `assets-originals/` and serve optimised copies through `next/image` (WebP or AVIF, responsive sizes).

Asset	Source file	Target path	Where used
Portrait	`Ezra.jpeg`	`public/images/ezra-portrait.jpg`	Hero, About, social share image
Hero visual	`WhatsApp_Image_2026-02-20_at_09_03_23.jpeg`	`public/images/hero-data-orb.jpg`	Hero background accent

Logo and brand images

Asset	Source file	Target path	Where used
Logo mark (head and swirl, purple to cyan)	`WhatsApp_Image_2026-10-09_at_13_45_50.jpeg`	`public/images/brand/logo-mark.png`	Header logo, favicon, app icon, social share image, footer
Neural brain illustration (blue line art)	`WhatsApp_Image_2026-10-09_at_13_47_49.jpeg`	`public/images/brand/ml-brain.png`	ML Course page, "Learn Machine Learning" offer card, section accents

Logo mark notes

- Circular composition of a stylised head in profile with swirling purple and cyan lines, on a dark background. Its colours match the Midnight Aurora theme.
- Cut it out from the dark square so it becomes a transparent circular logo. Export 512, 192, 180 and 32 px versions for the favicon and Apple touch icon, plus a 1200 by 630 social share card.

- Header: show it at about 40 px next to the name "Ezra Ndinya Ogechi". Footer: show it larger beside the tagline.
- On the Daylight theme, place it on a dark rounded tile so it keeps its contrast.
- It is about 800 by 800 px. Use it at 400 px or smaller. Ask Ezra for an SVG or larger file for any print use.

Brain illustration notes

- Half circuit-board, half brain, drawn in sky-blue line art on black. The artwork sits off-centre inside a larger black canvas, so crop tightly around the brain before use.
- Remove the black background so it sits cleanly on every theme. On Daylight, recolour it to indigo or place it on a dark tile.
- Use it as a decorative accent only (section accents only, because the NeuralNest logo is the course mark). It should never be the main hero visual.

Rights check: Ezra should confirm he owns or is licensed to use all of the brand images, since they look like generated or stock artwork, and keep a note of the source in the repo's README.

NeuralNest (course and learning brand)

Asset	Source file	Target path	Where used
NeuralNest logo (a brain above an open book, in teal, green and blue)	`NeuralNest.jpg`	`public/images/brand/neuralnest-logo.png`	`/course` header, `/learn` header, lesson page headers, the ML Course teaser and offer card, the Free Math Lessons teaser, the course social share image

- Name and line: **NeuralNest**, with "A welcoming home for learners exploring neural networks, AI, and machine learning."

- The image has no lettering, so the site must typeset the word NeuralNest next to the mark, in the heading font.
- The file is 640 by 640 px with the artwork in the middle and a wide plain dark-navy margin. Crop tightly around the brain and the book. Remove the navy background to make a transparent PNG, and keep a second copy with the navy background as a tile.
- On Midnight Aurora, Terminal and Ember the logo sits directly on the dark background. On Daylight, place it on a dark rounded tile so it keeps its contrast.
- Size: about 48 px high in headers and about 96 px in a page hero. Never stretch it.
- Alt text: "NeuralNest logo: a brain above an open book."
- The course page title is "NeuralNest ML Course by Ezra Ndinya Ogechi". Generate a 1200 by 630 share card for it from the logo and the line above.
- The same rights check applies: Ezra confirms he owns or is licensed to use the image.

Portrait notes

- Portrait orientation (about 3:4), indoor lighting, warm wall and blue curtain behind. Crop to head and shoulders and show it in a rounded frame.
- Remove or soften the background (a cut-out on a glowing ring works well) so it looks clean on all four themes. Keep a plain cropped version as a fallback.
- Alt text: "Ezra Ndinya Ogechi, data scientist and machine learning engineer."
- Generate a 1200 by 630 social share image from it with the name and role.

Hero visual notes

- The image shows a blue wireframe hand holding a glowing sphere with an atom and database symbol, surrounded by code, cloud, chart and chip icons.
- It is small (roughly 340 by 260 px), so do not stretch it full-width. Show it at no more than about 340 px wide or as a faded, blurred backdrop, and ask Ezra for a larger version if one exists.
- Its blue palette matches Midnight Aurora. For Daylight, Ember and Terminal, recolour it with a CSS hue shift or blend mode, or swap in a generated gradient orb in that theme's accent colour.
- Ezra should confirm he has the right to use this image (for example a licensed stock image) before launch.

16.1 Extra acceptance criteria

1. The Data Analytics section shows all seven analytics projects with working GitHub links and a visible **Visit my Data Analytics site** button.
2. The portrait and hero visual display sharply on mobile and desktop, with no stretching and no layout shift.
3. The hero image looks intentional in every theme.
4. Counters on the site match the real number of published projects, and no testimonials appear until real ones exist.
5. Tapping WhatsApp from any service card opens a chat with the matching prefilled message.

Logo and naming criteria

- The header shows the logo mark and "Ezra Ndinya Ogechi", and the browser tab shows the logo as the favicon.
- The ML course page and the Learn section use the NeuralNest name, line and logo, and the word NeuralNest is typeset beside the logo.
- Every page footer reads "© 2026 Ezra Ndinya Ogechi. All rights reserved."
- All six social links and the WhatsApp link open the correct pages in a new tab.

17. Chana and Certifications: Acceptance Criteria and Open Inputs

Acceptance criteria

1. The home page shows a Chana spotlight card directly after the Data Analytics Projects section, with the First Runner-Up badge.
2. `/projects/chana` exists, Chana is listed under Data Science with the AI Applications chip, and no invented features or numbers appear anywhere.
3. If the Chana repository is private or missing, the page still shows the GitHub button for the Chana address and also offers **Source code available on request** instead.
4. `/certifications` lists the Chana award and the GOMYCODE certificate in separate groups, each opening its file in a new tab.
5. The GOMYCODE entry is labelled "Participation" everywhere it appears, and the words "certified" and "qualification" are not used for it.
6. The Verify button appears only when a verification address is present in the content file.

7. The home page section order is: hero, Currently Building, About, What I Offer, Data Science Projects, Data Analytics Projects, Featured Work (Chana), Skills and Tools, Certifications and Recognition, ML Course teaser, Free Math Lessons teaser, Apps on Google Play (when present), contact call to action, footer.

Additional criteria (transcript and wording)

- The St. Ann's College transcript appears under Certificates with its three modules and grades, and the registration number is not shown as text or readable in the preview.
- The Certifications page contains no "verified" wording except beside a Verify button that has a real link.
- The page introduction makes no claim beyond the facts in the entries.

Inputs needed from Ezra for this addition

- Chana: the hackathon's name, organiser and date; proof of the placement (certificate, photo or announcement link); team members and Ezra's role; a short description of what Chana does, how it works (chatbot, navigation approach, tools) and what it achieved; screenshots or a demo video; a live demo link if one exists.
- Chana repository: make it public or confirm the correct address (see Section 5.5c).
- The correct official university name. This PRD uses "Dedan Kimathi University of Technology".
- GOMYCODE certificate: the PDF and a PNG export, plus the full verification address.
- Optional: whether project BIA from the GOMYCODE hackathon has a repository or demo worth featuring.

More inputs

- A clear scan or PDF of the St. Ann's College transcript, and the separate Certiport certificate or exam result if Ezra has one.
- A tidy export of the GOMYCODE certificate without the dark strip along the top of the current screenshot.

Risks

Risk	Mitigation
Chana repository is private, so the link looks broken	Make the repository public. Until then, show the on-request note beside the GitHub button

Risk	Mitigation
Overstating the GOMYCODE certificate	Fixed "Participation" label and the wording rules in Section 5.5d
An award claim visitors cannot check	Show a proof link as soon as Ezra provides one

18. Free Math Lessons (Math for Machine Learning)

Status: the structure below is final. The lesson content is written by the PRD author from Ezra's handwritten notes and is added to this section as Sections 18.6 to 18.10. The build agent implements the lessons exactly as written and must not add, reword or "improve" any mathematics.

18.1 Purpose

A free teaching section that explains the math needed for machine learning, in plain language, for complete beginners. It sits beside the ML course and brings in learners who then enrol. Topics, in this order:

1. Introduction: the big picture of math for machine learning (from Ezra's notes)
2. Linear algebra (from Ezra's notes, plus one lesson added by the author)
3. Statistics (from Ezra's notes, plus short additions)
4. Probability (from Ezra's notes, plus one short lesson added by the author)
5. Calculus, only the parts machine learning uses (written by the author, because the notes only introduce it)

Further maths is taught inside the ML course as models are built from scratch. The free section does not try to cover everything.

18.2 Routes and navigation

- `/learn`: overview with the five topics as cards, in order, each showing lesson count and estimated time.
- `/learn/math/[topic]/[lesson]`: one page per lesson.
- Add **Learn** to the main navigation and the footer. On the home page, add a **Free Math Lessons** teaser after the ML Course teaser, with a button to `/learn`.

- No sign-up and no payment. Every lesson has a **Join the ML course** call to action at the end.

Branding of the learning area: Show the NeuralNest logo, name and line at the top of `/learn` and `/course`, and a small NeuralNest mark in the header of every lesson page. The `/learn` page title is "NeuralNest: Free Math for Machine Learning". The rest of the site keeps the Ezra Ndinya Ogechi brand.

18.3 Lesson page layout (every lesson)

1. Title, topic, estimated time, and prerequisites (linked)
2. **Goal:** one sentence on what the learner will be able to do
3. **The idea in plain words**
4. **Worked examples**, step by step, each step on its own line
5. **Graph** wherever the lesson says so
6. **Key rules** summary box
7. **Practice questions** with answers hidden behind a Show answer button
8. **Where this is used in machine learning**
9. Previous and next lesson links, and the course call to action

18.4 Technical rules for the build

- Content lives in `content/learn/math/<topic>/<nn-slug>.md`, with front matter: `title`, `topic`, `order`, `summary`, `minutes`, `prerequisites`, `mlUse`.
- Render all equations with KaTeX at build time (`remark-math` and `rehype-katex`), so it works in the static export. No raw LaTeX may be visible on any page, and equations must scroll horizontally inside their own box on small screens.
- Graphs are written in the lesson file as a fenced block named `graph` containing JSON, exactly as defined in Section 18.5 (types `plot`, `scatter` and `bars`). Render `plot` and `scatter` in the browser with `function-plot`, and `bars` with a small custom SVG component, all loaded only on lesson pages. Every graph also shows its caption under it, which doubles as the accessible description.
- Each graph block in the lessons lists **checkpoints** (points the curve must pass through). The build agent must confirm each graph passes through its checkpoints, and fix the graph if not.
- Graphs follow the active theme (axis, grid and curve colours come from the theme tokens) and stay readable in all four themes.

- Show a progress marker per lesson, stored in the visitor's browser only.
- Lessons are free to read, and printable (clean print stylesheet).
- Acceptance: every lesson renders with correct equations and graphs on mobile and desktop, in all four themes, with no console errors, and lesson order matches Sections 18.6 to 18.10 exactly.

18.5 Lesson authoring format and graph specification

Lesson blocks. Every lesson in Sections 18.6 to 18.10 starts with a heading `#### Lesson N.N` and a title, followed by one line of metadata in code format. The build agent creates one file per lesson at the path in the `file` field. The front matter comes from the metadata (`title` , `topic` , `order` , `minutes` , `prerequisites`). The body is the lesson text copied exactly, from **Goal** to the last practice answer. Numbers, equations and words must be copied unchanged.

Math syntax. In this PRD every display formula sits in its own `latex` code block of bare TeX. In the lesson file, write each one between double dollar signs and render it with KaTeX. Short inline maths is plain text with Unicode symbols (θ, √, ×, ·, ², ⁻¹, ≈). Copy inline maths as plain text and do not convert it.

Practice format. Each question and each answer is its own bullet. Questions start with `Q1.` , `Q2.` and so on, and answers start with `A1.` , `A2.` . The build turns every answer bullet into a hidden answer behind a **Show answer** button.

Callouts. A paragraph starting with **Key rules** becomes the Key rules box. A paragraph starting with **Where this is used in machine learning** becomes the ML box. A paragraph starting with **Note** becomes a Note box.

Code. Python blocks are shown with a Copy button and are never executed.

Graph block. A fenced block named `graph` containing JSON. Keys that every graph has: `type` (`plot` , `scatter` or `bars`), `caption` (required), and `checkpoints` . Colours are theme tokens: `accent` , `accent2` , `muted` , `text` .

- `plot` : `xDomain` , `yDomain` , and any of `functions` (items with `fn` , `label` , `color`), `vectors` (items with `from` , `to` , `label` , `color` , `dashed`), `points` (items with `at` , `label`), `areas` (items with `fn` , `from` , `to`) and `vlines` (items with `x` , `label`).
- `scatter` : `xDomain` , `yDomain` , `xLabel` , `yLabel` , `points` as a list of [x, y] pairs, and an optional `line` with `slope` and `intercept` .
- `bars` : `xLabel` , `yLabel` , `labels` , and `series` (items with `name` , `values` , `color`). An optional `style` of `lines` draws connected points instead of bars.

- Function expressions use plain math such as `x^2` or `exp(-x^2/2)`. Never use `pi`, write the number instead.
- `checkpoints` lists [x, y] points a curve or vector must pass through (for `bars`, [label, value] pairs, with a third item naming the series when there are several). The build agent must check every checkpoint after drawing and fix the graph if any is off by more than 1 percent of the axis range.

18.6 Topic 0: Introduction

Lesson 0.1 Mathematics for Machine Learning: The Big Picture

`file: content/learn/math/00-introduction/01-big-picture.md` `minutes: 6` `prerequisites: none`

Goal: Name the four areas of math used in machine learning and say what each one is for.

The idea in plain words Every machine learning system follows one path: **Data, then Model, then Training**.

- **Data** is information turned into numbers, such as the size, age and number of bedrooms of a house.
- A **model** is a rule that turns data into an answer, such as a predicted price.
- **Training** means adjusting the model until its answers are close to the true ones.

Note The build agent shows these three steps as three cards joined by arrows.

Math is the language of every step. Four areas do the work:

Area	The question it answers	What you will meet
Linear algebra	How do we store data and combine it?	Vectors such as [2, -8, 7], and matrices with rows and columns
Statistics	How do we describe data and learn from a sample?	Mean, median, mode, standard deviation, descriptive and inferential statistics
Probability	How likely is something to happen?	How likely is it that a person has diabetes or not?
Calculus	How does a result change when an input changes, and how do we improve it?	Differential calculus, integral calculus and gradient descent

Calculus in one minute. There are two types. Differential calculus splits something into very small pieces and looks at how it changes. Integral calculus adds many small pieces back into one large whole. Gradient descent, the method that trains most models, is built from differential calculus. Lesson 4.4 builds it from scratch.

Key rules

- Data, then Model, then Training is the path every ML project follows.
- Linear algebra stores and combines data. Statistics describes it. Probability measures uncertainty. Calculus improves the model.

Practice

- Q1. Put these in order: Training, Data, Model.
- A1. Data, Model, Training.
- Q2. Which area of math helps you ask how likely it is that a person has diabetes?
- A2. Probability.
- Q3. Which kind of calculus adds many small pieces into one large whole?
- A3. Integral calculus.

Where this is used in machine learning: This lesson is the map. Every later lesson is one piece of it.

18.7 Topic 1: Linear Algebra

Linear algebra is the math of lists and tables of numbers. A list is a vector and a table is a matrix. Almost every ML model stores its data in one of these two shapes.

Lesson 1.1 Vectors: Three Ways to See Them

`file: content/learn/math/01-linear-algebra/01-vectors.md` `minutes: 8` `prerequisites: 0.1`

Goal: Explain what a vector is from the physics, mathematics and computer science points of view, and tell a scalar from a vector.

The idea in plain words A vector can be seen in three ways. All three describe the same thing.

1. The physics view. Some quantities have only a size. Speed is one: 60 km/h is a **scalar**. Some quantities have a size and a direction. Velocity is one: 60 km/h north is a **vector**. A scalar is simply one number.

2. The mathematics view. A vector is an arrow drawn on a graph. It has a **magnitude** (how long the arrow is) and a **direction** (where it points). The next lesson draws these.

3. The computer science view. A vector is a list of numbers, like an array in Python. It can be written as a row or as a column:

```
[\,2,\ -8,\ 7\,] \qquad \begin{bmatrix} 2 \\ -8 \\ 7 \end{bmatrix}
```

Worked example. Work experience and salary. Each person becomes one vector of two numbers, [years of experience, salary]. One person is [5, 50000] and another is [10, 100000].

```
import numpy as np
person_1 = np.array([5, 50000])
person_2 = np.array([10, 100000])
print(person_1)
```

Key rules

- A scalar is a single number. A vector is a list of numbers, or an arrow with size and direction.
- A vector can be a row or a column. Its numbers are called components.
- The number of components is the dimension. [2, -8, 7] has 3 components.

Practice

- Q1. Is a temperature of 25 degrees Celsius a scalar or a vector?
- A1. A scalar, because it has size only.
- Q2. Is a wind of 20 km/h blowing east a scalar or a vector?
- A2. A vector, because it has size and direction.
- Q3. How many components does [5, 50000] have?
- A3. Two.

Where this is used in machine learning: Data is stored as vectors. One house can be the vector [1450, 3, 12] for area, bedrooms and age. A whole dataset is many such vectors stacked together, which is a matrix (Lesson 1.6).

Lesson 1.2 Vectors on a Graph: Magnitude and Direction

`file: content/learn/math/01-linear-algebra/02-magnitude-direction.md` `minutes: 10` `prerequisites: 1.1`

Goal: Draw a 2D vector, then calculate its magnitude and its direction.

The idea in plain words Every vector has two properties: **magnitude** (its length) and **direction** (the angle it points at). On a graph, a vector starts at the origin and ends at the point given by its components. The vector (2, 2) goes 2 steps right and 2 steps up. It can also be written 2i + 2j, where i = (1, 0) and j = (0, 1) are one-step arrows along the x-axis and y-axis.

```
{"type":"plot","xDomain":[-4,4],"yDomain":[-1,4],"vectors":[{"from":
[0,0],"to":[2,2],"label":"(2, 2)","color":"accent"},{"from":[0,0],"to":
[-3,2],"label":"(-3, 2)","color":"accent2"}],"checkpoints":[[2,2],
[-3,2]],"caption":"Two vectors drawn from the origin. (2, 2) points up and to
the right. (-3, 2) points up and to the left."}
```

Step 1: Magnitude. Use Pythagoras. For a vector with components a1 and a2:

```
|v| = \sqrt{a_1^2 + a_2^2}
```

For (2, 2): $|v| = \sqrt{(2^2 + 2^2)} = \sqrt{8} \approx 2.83$.

Step 2: Direction. Draw the right-angled triangle under the vector. The side along the x-axis (adjacent) is 2, the side going up (opposite) is 2, and the vector itself is the hypotenuse. Remember SOH CAH TOA:

```
\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \cos\theta =
\frac{\text{adjacent}}{\text{hypotenuse}} \qquad \tan\theta =
\frac{\text{opposite}}{\text{adjacent}}
```

Here tan θ = 2 / 2 = 1, so θ = tan⁻¹(1) = 45°. The direction of (2, 2) is 45 degrees.

A second example. For (-3, 2), the magnitude is √((-3)² + 2²) = √13 ≈ 3.61. The triangle angle is tan⁻¹(2 / 3) ≈ 33.69°. Because the arrow points up and to the left, its direction measured from the positive x-axis is 180° - 33.69° = 146.31°.

Key rules

- Magnitude: |v| = √(a1² + a2²).
- Direction: θ = tan⁻¹(a2 / a1), adjusted for the quadrant the arrow is in.
- A vector is fully described by its magnitude and direction, or by its components.

Practice

- Q1. What is the magnitude of (3, 4)?
- A1. √(9 + 16) = √25 = 5.
- Q2. What is the direction of (1, 1)?
- A2. tan θ = 1, so 45 degrees.
- Q3. What is the magnitude of (0, 5), and which way does it point?
- A3. The magnitude is 5. It points straight up, at 90 degrees.

Where this is used in machine learning: The magnitude of a data vector measures its distance from the origin. Distance is the heart of methods that group similar data points, such as clustering.

Lesson 1.3 Vector Operations: Add, Subtract and Scale

`file: content/learn/math/01-linear-algebra/03-vector-operations.md` `minutes: 10` `prerequisites: 1.2`

Goal: Add, subtract and scale vectors, and see what each does on a graph.

The idea in plain words Three operations change vectors: adding two vectors, subtracting two vectors, and multiplying a vector by a scalar (a single number). Adding or subtracting two vectors always gives a vector.

Addition. Add the matching components. For a = (2, 1) and b = (1, 3):

a + b = (2 + 1, 1 + 3) = (3, 4)

On the graph, place the tail of b at the tip of a. The arrow from the origin to the end of b is a + b.

```
{"type":"plot","xDomain":[-1,5],"yDomain":[-1,5],"vectors":[{"from":
[0,0],"to":[2,1],"label":"a = (2, 1)","color":"accent"},{"from":[0,0],"to":
[1,3],"label":"b = (1, 3)","color":"accent2"},{"from":[2,1],"to":
[3,4],"label":"b moved to the tip of a","color":"accent2","dashed":true},
{"from":[0,0],"to":[3,4],"label":"a + b = (3,
4)","color":"text"}],"checkpoints":[[2,1],[1,3],[3,4]],"caption":"Adding
vectors tip to tail. Vector a is (2, 1), vector b is (1, 3), and their sum is
(3, 4)."}
```

Subtraction. Subtract the matching components. Subtracting b is the same as adding the opposite arrow: a - b = a + (-b). Example:

(5, 3) - (3, 2) = (5 - 3, 3 - 2) = (2, 1)

Multiplying a vector by a scalar. Multiply every component by the number. Take v = (3, 3):

- 2 × v = (6, 6). Same direction, twice as long.
- 0.5 × v = (1.5, 1.5). Same direction, half as long.
- -0.5 × v = (-1.5, -1.5). Half as long, and the direction is flipped.

```
import numpy as np
a = np.array([2, 1])
b = np.array([1, 3])
print(a + b)      # [3 4]
print(a - b)      # [ 1 -2]
print(-0.5 * np.array([3, 3]))   # [-1.5 -1.5]
```

Key rules

- Add or subtract two vectors by combining matching components. Both vectors must have the same number of components.
- A scalar larger than 1 stretches a vector. A scalar between 0 and 1 shrinks it. A negative scalar flips its direction.

Practice

- Q1. Work out (4, 1) + (2, 5).
- A1. (6, 6).
- Q2. Work out (7, 4) - (3, 4).
- A2. (4, 0).

- Q3. Work out 3 × (1, -2).
- A3. (3, -6).

Where this is used in machine learning: Models combine data by adding vectors and rescale it by multiplying by numbers. Training itself is a long series of small vector additions, as Lesson 4.4 shows.

Lesson 1.4 Angle Between Vectors and the Dot Product

`file: content/learn/math/01-linear-algebra/04-angle-dot-product.md` `minutes: 12` `prerequisites: 1.3`

Goal: Calculate a dot product and use it to find the angle between two vectors.

The idea in plain words When the angle between two vectors is very small, the vectors point almost the same way, so they are similar. This idea is called **cosine similarity**. Data points that sit close together in direction share similar properties.

The dot product. Multiply the matching components and add the results. The answer is a **scalar**, not a vector. For a = (2, 3) and b = (4, 4):

a · b = (2 × 4) + (3 × 4) = 8 + 12 = 20

Finding the angle. The dot product is also equal to

```
a \cdot b = |a|\,|b|\cos\theta
```

where θ is the angle between the two vectors. Work through it:

- |a| = √(2² + 3²) = √13 ≈ 3.606
- |b| = √(4² + 4²) = √32 ≈ 5.657
- cos θ = 20 / (3.606 × 5.657) = 20 / 20.396 ≈ 0.9806
- θ = cos⁻¹(0.9806) ≈ 11.31°

The angle is small, so a and b are very similar in direction.


```
{"type":"plot","xDomain":[-1,5],"yDomain":[-1,5],"vectors":[{"from":
[0,0],"to":[2,3],"label":"a = (2, 3)","color":"accent"},{"from":[0,0],"to":
[4,4],"label":"b = (4, 4)","color":"accent2"}],"checkpoints":[[2,3],
[4,4]],"caption":"Vectors a = (2, 3) and b = (4, 4) from the origin. The
angle between them is about 11.31 degrees."}
```

```
import numpy as np
a = np.array([2, 3])
b = np.array([4, 4])
print(np.dot(a, b))                       # 20, a single value
cos_theta = np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))
print(np.degrees(np.arccos(cos_theta)))  # about 11.31
```

To draw a vector with matplotlib use `plt.quiver(0, 0, x, y, angles='xy', scale_units='xy', scale=1)`.

Key rules

- Dot product: multiply matching components, then add. The result is a scalar.
- a · b = |a| |b| cos θ.
- Cosine similarity is cos θ. It is 1 when the vectors point the same way, 0 when they are at 90 degrees, and -1 when they point opposite ways.

Practice

- Q1. Work out (1, 2) · (3, 4).
- A1. 1 × 3 + 2 × 4 = 11.
- Q2. Work out (1, 0) · (0, 1). What does the answer say about the angle?
- A2. It is 0, so the angle is 90 degrees.
- Q3. If the angle between two vectors is very small, are they similar or different?
- A3. Similar.

Where this is used in machine learning: Cosine similarity compares two items by direction. Recommendation systems, such as a movie recommender, and text-matching tools often rely on it.

Lesson 1.5 Cross Product and Projection

`file: content/learn/math/01-linear-algebra/05-cross-product-projection.md` `minutes: 12` `prerequisites: 1.4`

Goal: Calculate a cross product and a projection, and say what kind of answer each one gives.

The idea in plain words There are two ways to multiply vectors. The dot product gives a scalar. The **cross product** gives a **vector**. For two 3D vectors, the cross product is a new vector that stands at right angles to both of them, and its length is |a| |b| sin θ.

```
a \times b = |a|\,|b|\sin\theta\;\hat{n}
```

To calculate it, use a determinant with the unit vectors i, j and k:

```
a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3
\end{vmatrix} = i(a_2 b_3 - a_3 b_2) - j(a_1 b_3 - a_3 b_1) + k(a_1 b_2 - a_2
b_1)
```

Worked example. Take the same pair as the last lesson, with a third component of 0: a = (2, 3, 0) and b = (4, 4, 0).

```
\begin{vmatrix} i & j & k \\ 2 & 3 & 0 \\ 4 & 4 & 0 \end{vmatrix} = i(3 \cdot
0 - 0 \cdot 4) - j(2 \cdot 0 - 0 \cdot 4) + k(2 \cdot 4 - 3 \cdot 4) = -4k
```

The answer is the vector (0, 0, -4). It points along the z-axis, at right angles to the flat page where a and b lie. Check its length: |a| |b| sin θ = 3.606 × 5.657 × sin 11.31° ≈ 4, which matches.

```
import numpy as np
a = np.array([2, 3, 0])
b = np.array([4, 4, 0])
print(np.cross(a, b))   # [ 0  0 -4]
```

Projection of a vector. The projection of a on b is the shadow that a casts onto the direction of b. The result is a **vector** that points along b.

```
\text{proj}_{b}\,a = \frac{a \cdot b}{|b|^2}\; b
```

For a = (2, 3) and b = (4, 4): a · b = 20 and |b|² = 32, so the factor is 20 / 32 = 0.625. The projection is 0.625 × (4, 4) = (2.5, 2.5).

```
{"type":"plot","xDomain":[-1,5],"yDomain":[-1,5],"vectors":[{"from":
[0,0],"to":[2,3],"label":"a = (2, 3)","color":"accent"},{"from":[0,0],"to":
[4,4],"label":"b = (4, 4)","color":"accent2"},{"from":[0,0],"to":
[2.5,2.5],"label":"projection of a on b = (2.5, 2.5)","color":"text"},
{"from":[2,3],"to":
[2.5,2.5],"label":"","color":"muted","dashed":true}],"checkpoints":[[2,3],
[4,4],[2.5,2.5]],"caption":"The shadow of a = (2, 3) on b = (4, 4) is the
vector (2.5, 2.5). The dashed line from the tip of a meets b at a right
angle."}
```

Key rules

- Dot product gives a scalar. Cross product gives a vector.
- The cross product of two 3D vectors is at right angles to both.
- Projection of a on b = (a · b / |b|²) × b, and it is a vector along b.

Practice

- Q1. Which product gives a single number, the dot product or the cross product?
- A1. The dot product.
- Q2. Find the projection of a = (3, 4) on b = (1, 0).
- A2. a · b = 3 and |b|² = 1, so the projection is 3 × (1, 0) = (3, 0).
- Q3. Find the cross product of (1, 0, 0) and (0, 1, 0).
- A3. i(0·0 - 0·1) - j(1·0 - 0·0) + k(1·1 - 0·0) = k, which is (0, 0, 1).

Where this is used in machine learning: Projection is the idea behind finding a best-fit line and behind squeezing many features into fewer ones. The course returns to it when models are built from scratch.

Lesson 1.6 Matrices: Scalar, Vector, Matrix and Shape

`file: content/learn/math/01-linear-algebra/06-matrices-shape.md` `minutes: 10` `prerequisites: 1.1`

Goal: Tell a scalar, a vector and a matrix apart, and read the shape of a matrix.

The idea in plain words There are three kinds of data holders. A **scalar** is a single number, such as 24. A **vector** is a list of numbers, either a row or a column. A **matrix** is a table with several rows and several columns.

```
[\,2,\ -8,\ 7\,] \qquad \begin{bmatrix} -4 \\ -4 \\ 27 \end{bmatrix} \qquad
\begin{bmatrix} 1 & 4 & 5 \\ 2 & 7 & 3 \\ 8 & -3 & 1 \end{bmatrix}
```

The first is a row vector, the second is a column vector and the third is a matrix.

Shape of a matrix. The shape is written rows × columns. Read it as: number of rows first, then number of columns.

```
\begin{bmatrix} 2 & 5 \\ 4 & 8 \end{bmatrix} \text{ is } 2 \times 2 \qquad
\begin{bmatrix} 8 & 6 & 1 \\ 2 & 9 & 2 \\ 3 & 4 & 3 \end{bmatrix} \text{ is }
3 \times 3 \qquad \begin{bmatrix} 2 & 5 \\ 4 & 8 \\ 6 & 1 \end{bmatrix}
\text{ is } 3 \times 2
```

A matrix does not have to be square. It can have a different number of rows and columns. A vector is a special matrix with only one row or only one column.

General notation. The entry in row i and column j is written a(i, j). A matrix with m rows and n columns is an m × n matrix.

```
\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}
```

```
import numpy as np
M = np.array([[1, 4, 5], [2, 7, 3], [8, -3, 1]])
print(M.shape)                  # (3, 3)
R = np.random.rand(3, 3)        # a random 3 x 3 matrix
```

Key rules

- Scalar: one number. Vector: one row or one column. Matrix: rows and columns.
- Shape = rows × columns.
- In a(i, j), i is the row and j is the column.

Practice

- Q1. What is the shape of the matrix with rows [1, 4, 5] and [2, 7, 3]?
- A1. 2 × 3, because it has 2 rows and 3 columns.
- Q2. In a(2, 3), which row and which column is meant?
- A2. Row 2 and column 3.
- Q3. The vector [2, -8, 7] is also a matrix. What is its shape?
- A3. 1 × 3, one row and three columns.

Where this is used in machine learning: A dataset is a matrix. Every row is one record, such as one house, and every column is one feature, such as area or age. In pandas, `house_data.head()` shows the first 5 rows of that matrix.

Lesson 1.7 Special Matrices and the Transpose

`file: content/learn/math/01-linear-algebra/07-special-matrices-transpose.md` `minutes: 8` `prerequisites: 1.6`

Goal: Recognise a null matrix and an identity matrix, and flip a matrix with the transpose.

The idea in plain words Null matrix (zero matrix). Every entry is 0. It can be any shape.

```
\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
```

Identity matrix. All the values on the main diagonal (top left to bottom right) are 1 and every other value is 0. It is always square.

```
\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
```

Transpose. The transpose of a matrix is formed by turning all its rows into columns, and the columns into rows. It is written with a small T.

```
A = \begin{bmatrix} 2 & 5 \\ 4 & 8 \end{bmatrix} \qquad A^T = \begin{bmatrix} 2 & 4 \\ 5 & 8 \end{bmatrix}
```

```
B = \begin{bmatrix} 8 & 6 & 1 \\ 2 & 9 & 2 \\ 3 & 4 & 3 \end{bmatrix} \qquad B^T = \begin{bmatrix} 8 & 2 & 3 \\ 6 & 9 & 4 \\ 1 & 2 & 3 \end{bmatrix}
```

If A is m × n, then its transpose is n × m.

```
import numpy as np
print(np.zeros((2, 2)))   # null matrix
print(np.eye(3))          # 3 x 3 identity matrix
A = np.array([[2, 5], [4, 8]])
print(A.T)                # [[2 4] [5 8]]
```

Key rules

- Null matrix: all zeros. Identity matrix: ones on the main diagonal, zeros elsewhere.
- Transpose: rows become columns and columns become rows. An m × n matrix becomes n × m.

Practice

- Q1. Write the transpose of the row vector [1, 2, 3].
- A1. The column with entries 1, 2 and 3 (a 3 × 1 matrix).
- Q2. If A has shape 3 × 2, what is the shape of its transpose?
- A2. 2 × 3.
- Q3. Write the 2 × 2 identity matrix.
- A3. The matrix with rows [1, 0] and [0, 1].

Where this is used in machine learning: Transposes are used to line up shapes before multiplying matrices. The identity matrix acts like the number 1: it leaves a matrix unchanged when you multiply by it.

Lesson 1.8 Matrix Operations: Add, Subtract, Scale and Multiply

`file: content/learn/math/01-linear-algebra/08-matrix-operations.md` `minutes: 15` `prerequisites: 1.7`

Goal: Add, subtract, scale and multiply matrices, and check whether an operation is allowed before doing it.

The idea in plain words There are four matrix operations. Two of them have a strict rule about shape.

1. Addition. Two matrices can be added only if they have the same shape, meaning the same number of rows and the same number of columns. Add matching entries.

```
\begin{bmatrix} 2 & 3 \\ 10 & 5 \end{bmatrix} + \begin{bmatrix} 10 & 5 \\ 20 & 4 \end{bmatrix} = \begin{bmatrix} 12 & 8 \\ 30 & 9 \end{bmatrix} \qquad (2 \times 2) + (2 \times 2) = (2 \times 2)
```

```
\begin{bmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ 3 & 6 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 7 & 8 \\ 8 & 8 \end{bmatrix} \qquad (3 \times 2) + (3 \times 2) = (3 \times 2)
```

2. Subtraction. The same rule: both matrices must have the same shape. Subtract matching entries.

```
\begin{bmatrix} 2 & 3 \\ 10 & 5 \end{bmatrix} - \begin{bmatrix} 10 & 5 \\ 20 & 4 \end{bmatrix} = \begin{bmatrix} -8 & -2 \\ -10 & 1 \end{bmatrix}
```

3. Multiplying by a scalar. Multiply every entry by the number. This works for vectors too, because a vector is a matrix with one row or one column.

```
5 \times \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix} \qquad 5 \times \begin{bmatrix} 2 & 1 \\ 4 & 2 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 20 & 10 \\ 30 & 15 \end{bmatrix}
```

4. Multiplying two matrices. The rule: the number of columns in the first matrix must equal the number of rows in the second. The answer has the same number of rows as the first matrix and the same number of columns as the second.

```
(m \times n)\,(n \times p) = (m \times p)
```

For 2 × 2 matrices the general pattern is:

```
\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h
\end{bmatrix} = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh
\end{bmatrix}
```

Each entry is a dot product of a row from the first matrix with a column from the second. Worked example:

```
\begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 10 & 5 \\ 2 & 4
\end{bmatrix} = \begin{bmatrix} 2 \cdot 10 + 3 \cdot 2 & 2 \cdot 5 + 3 \cdot
4 \\ 1 \cdot 10 + 5 \cdot 2 & 1 \cdot 5 + 5 \cdot 4 \end{bmatrix} =
\begin{bmatrix} 26 & 22 \\ 20 & 25 \end{bmatrix}
```

When it is not allowed. A 3 × 2 matrix times another 3 × 2 matrix cannot be multiplied, because the first has 2 columns and the second has 3 rows, and 2 is not equal to 3.

Order matters. Multiplying in the other order gives a different answer. Using the same two matrices, the second one times the first gives the matrix with rows [25, 55] and [8, 26], which is not the same as [26, 22] and [20, 25].

```
import numpy as np
A = np.array([[2, 3], [1, 5]])
B = np.array([[10, 5], [2, 4]])
print(A + B)
print(A - B)
print(5 * A)
print(A @ B)     # [[26 22] [20 25]]
print(B @ A)     # [[25 55] [ 8 26]]
```

Key rules

- Add or subtract: the two matrices must have the same shape.
- Scalar multiplication: multiply every entry.
- Matrix multiplication: columns of the first must equal rows of the second. The result has the rows of the first and the columns of the second.
- A × B is usually not the same as B × A.

Practice

- Q1. Can a 2 × 3 matrix be added to a 3 × 2 matrix?
- A1. No, because the shapes are different.
- Q2. A is 2 × 3 and B is 3 × 4. What is the shape of A × B?
- A2. 2 × 4.
- Q3. Work out 3 times the matrix with rows [1, 2] and [0, -1].
- A3. The matrix with rows [3, 6] and [0, -3].

Where this is used in machine learning: Every prediction a model makes is a matrix multiplication. Lesson 1.9 shows it.

Lesson 1.9 Putting It Together: Predictions as a Matrix Times a Vector

`file: content/learn/math/01-linear-algebra/09-predictions-matrix-vector.md` `minutes: 8` `prerequisites: 1.8`

Goal: See how one matrix multiplication produces predictions for many examples at once.

The idea in plain words Suppose a simple model predicts a house price (in thousands) as 10 + 2 × size, where size is in square metres. The numbers 10 and 2 are the model's **weights**, written as the vector w = (10, 2). Put each house into a row of a matrix X. The first column is always 1, so the 10 acts as a starting price. The second column is the size. Two houses, of 50 and 80 square metres, give:

```
X w = \begin{bmatrix} 1 & 50 \\ 1 & 80 \end{bmatrix} \begin{bmatrix} 10 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 10 + 50 \cdot 2 \\ 1 \cdot 10 + 80 \cdot 2 \end{bmatrix} = \begin{bmatrix} 110 \\ 170 \end{bmatrix}
```

The shapes fit the rule: (2 × 2) times (2 × 1) gives (2 × 1), one predicted price per house. Each answer is the dot product of one house's vector with the weights.

```
import numpy as np
X = np.array([[1, 50], [1, 80]])
w = np.array([10, 2])
print(X @ w)   # [110 170]
```

Key rules

- Predictions for many examples at once = X × w.
- Each row of X is one example. The vector w holds the weights the model has learned.
- Training means finding the weights w that make the predictions close to the true values. Lesson 4.4 shows how.

Practice

- Q1. X has shape 3 × 2 and w has shape 2 × 1. What shape is the result?
- A1. 3 × 1, one prediction for each of the 3 examples.
- Q2. Use the same weights for a house of 100 square metres. What is the predicted price?
- A2. 10 + 2 × 100 = 210.
- Q3. Why does the first column of X contain only ones?
- A3. So that the first weight, 10, is added to every prediction as a starting value.

Where this is used in machine learning: This is the core of linear regression, including a house price predictor like the one in the portfolio.

18.8 Topic 2: Statistics

Statistics is the science of learning from data. This topic follows Ezra's notes in order: what statistics is, types of data, types of statistics and studies, sampling, central tendency, variability, percentiles, correlation, and hypothesis testing.

Lesson 2.1 What Is Statistics and Why Machine Learning Needs It

`file: content/learn/math/02-statistics/01-what-is-statistics.md` `minutes: 8` `prerequisites: 0.1`

Goal: Define statistics, name the main statistical measures, and give real uses of statistics.

The idea in plain words Statistics is the science concerned with developing and studying methods for collecting, analysing, interpreting and presenting data. In short, it is a tool that helps us extract information and knowledge from data.

The main statistical measures

- **Range:** the gap between the largest and the smallest value.
- **Mean:** the average value.
- **Standard deviation:** how far the values spread away from the mean.
- **Correlation:** how two things move together, for example work experience and salary.

Example. Who is the best footballer in the world between 2010 and 2020? Statistics answers this by collecting every player's results and comparing their **average**, which is the mean.

Applications of statistics

- **Six Sigma.** A data-driven business technique that improves quality by removing the causes of defects and reducing variation in a process. Statistics about a company's products help it work more efficiently.
- **Weather forecasting.** Collect information about the weather and climate at a given time of the year, then use it to predict the weather.

Role of statistics in machine learning. The path is Data, then Machine Learning models. Statistics prepares, summarises and checks the data before a model learns from it.

Key rules

- Statistics collects, analyses, interprets and presents data.
- The core measures are range, mean, standard deviation and correlation.

Practice

- Q1. In one sentence, what is statistics?
- A1. The science of collecting, analysing, interpreting and presenting data to get information from it.
- Q2. Which measure tells you how spread out the values are around the mean?
- A2. The standard deviation.
- Q3. What does Six Sigma try to reduce?
- A3. Defects and variation in a process.

Where this is used in machine learning: Before any model is trained, statistics is used to understand the data: its averages, its spread and how its features relate to each other.

Lesson 2.2 Types of Data

`file: content/learn/math/02-statistics/02-types-of-data.md` `minutes: 8` `prerequisites: 2.1`

Goal: Classify any column of data as nominal, ordinal, discrete or continuous.

The idea in plain words Data comes in two main kinds, and each splits in two:

Note The build agent draws this as a tree. The top box is Data. Under it are Categorical and Numerical. Under Categorical are Nominal and Ordinal. Under Numerical are Discrete and Continuous.

Kind	Type	Meaning	Examples
Categorical	Nominal	Labels with no natural order. Used purely for naming or labelling	Eye colour (blue, brown), marital status (single, married), blood type (A, B, AB, O), male or female, ball colours
Categorical	Ordinal	Labels that follow a natural order or ranking	Rating of a phone's quality, satisfaction level, education level, competition results (1st, 2nd, 3rd), socio-economic status
Numerical	Discrete	Numbers that can take only certain separate values	Number of children, number of goals
Numerical	Continuous	Numbers that can take almost any value, including decimals	Height (167.8 cm), temperature (36.6 degrees)

The main difference between nominal and ordinal data is that ordinal data has a natural order or ranking, and nominal data does not. Blue and brown eyes cannot be ranked. Satisfaction levels such as poor, fair and good can.

Discrete versus continuous. Discrete data can only take certain values. Continuous data can take almost any numeric value and, unlike discrete data, can have decimal values.

Key rules

- Categorical data names groups. Numerical data measures or counts.
- Nominal has no order. Ordinal has an order.
- Discrete has separate values. Continuous has any value in a range.

Practice

- Q1. What type of data is blood group?
- A1. Categorical, nominal.
- Q2. What type of data is education level (primary, secondary, university)?
- A2. Categorical, ordinal, because the levels have a natural order.
- Q3. What type of data is a weight of 62.5 kg?
- A3. Numerical, continuous.

Where this is used in machine learning: The type of a column decides how it is prepared. Labels are converted to numbers, and continuous numbers are often rescaled, before a model can use them.

Lesson 2.3 Descriptive and Inferential Statistics, and Types of Study

`file: content/learn/math/02-statistics/03-descriptive-inferential-studies.md` `minutes: 10` `prerequisites: 2.2`

Goal: Tell descriptive statistics from inferential statistics, and name the three types of statistical study.

The idea in plain words There are two types of statistics.

	Descriptive statistics	Inferential statistics
What it does	Describes the data	Gets inferences about the data
Related to	Data analysis	Data science
How	Gives simple summaries of a sample, using measures such as mean, median and mode	Draws data from a sample, then makes inferences and predictions about the larger population the sample came from
Main tools	Measures of central tendency (mean, median, mode) and measures of variability (range, standard deviation, variance)	Correlation between variables, shown for example as a heatmap

Types of statistical study. There are three.

- **Sample study:** a study carried out on a sample that represents the whole population. Example: find the average blood sugar level of a population by measuring a sample.
- **Observational study:** we only collect and analyse data. We do not inject any change. We just observe and look for relationships in the data.
- **Experimental study:** a study in which the variables are controlled and manipulated by the experimenter.

Observational example. Record people's physical activity and their body weight. More activity goes with lower weight. That is a **negative correlation**. Nothing was changed, we only observed.

{"type":"scatter","xDomain":[0,8],"yDomain":[60,100],"xLabel":"Physical activity (hours per week)","yLabel":"Body weight (kg)","points":[[1,95],[2,92],[3,88],[4,85],[5,80],[6,78],[7,72]],"line":{"slope":-3.75,"intercept":99.29},"checkpoints":[[1,95],[4,85],[7,72]],"caption":"Body weight falls as weekly physical activity rises. The points slope downwards, which is a negative correlation."}

Experimental example. Split people into two groups. Group 1 follows healthy habits and does regular activity. Group 2 does not and is the **control group**. The goal is to find the optimum blood sugar level. A clinical trial of a medicine works the same way: one group receives the medicine and the control group does not.

Key rules

- Descriptive statistics describes a dataset. Inferential statistics draws conclusions about a population from a sample.
- Observational study: observe only. Experimental study: the experimenter changes the variable and uses a control group.

Practice

- Q1. Which type of statistics predicts something about a whole population from a sample?
- A1. Inferential statistics.
- Q2. A researcher records the weight and the daily steps of 200 people without changing anything. What type of study is this?
- A2. An observational study.
- Q3. Why does an experimental study need a control group?
- A3. To compare against people who did not get the change, so the effect can be seen.

Where this is used in machine learning: Describing a dataset is descriptive statistics. Training a model on a sample and expecting it to work on new data is inferential statistics.

Lesson 2.4 Population, Sample and Sampling Techniques

`file: content/learn/math/02-statistics/04-sampling.md` `minutes: 12` `prerequisites: 2.3`

Goal: Explain population and sample, and choose between the four probability sampling techniques.

The idea in plain words The **population** is the whole group you want to know about. A **sample** is a smaller part of it, chosen to represent it. To find the average blood sugar level of a whole population, you measure a sample and use it to estimate the answer.

Sampling techniques come in two families. In **probability sampling**, the members of the population have an equal chance of being chosen. In **non-probability sampling** they do not. The four probability techniques are:

1. Simple random sampling. The sample is picked at random from a large population, so every data point has an equal probability of being selected. Example: an employee survey in a company.

- Pros: no sample bias, an unbiased sample, a simple method, and it needs less domain knowledge.
- Cons: the population size should be large, and it sometimes cannot represent the population well.

2. Systematic sampling. The sample is picked from the population at regular intervals. It is used when the population is homogeneous and the data points are uniformly distributed. Example: select every 10th member of a population of 10,000. That gives a sample of 1,000.

- Pros: quick and easy, less bias, even distribution of data.
- Cons: risk of data manipulation, it needs randomness in the data, and the population should not have patterns.

3. Stratified random sampling. The population is divided into smaller groups called **strata**, and samples are taken randomly from all of them. Example: smartphone sales in all the states.

- Pros: it captures the important characteristics of the population, and precision is high when the strata differ a lot from each other.
- Cons: it cannot be used on a population that cannot be sorted into groups, and data points may overlap between groups.

4. Cluster sampling. It is used when the population has inherent groups. The population is divided into **clusters**, and then random clusters are taken as the sample. Example: smartphone sales in randomly selected states.

- Pros: it needs fewer resources and reduces variability.
- Cons: it cannot be used on a population without natural groups, data points may overlap, and it may give only a general insight into the whole population.

Stratified versus cluster. Stratified sampling takes some members from every group. Cluster sampling takes whole groups, but only some of them.

Key rules

- Population is the whole. A sample is the part used to represent it.
- Probability sampling gives every member an equal chance of selection.
- Four techniques: simple random, systematic, stratified, cluster.

Practice

- Q1. Which technique selects every 10th member of a list?
- A1. Systematic sampling.
- Q2. A population of 10,000 is sampled systematically at every 10th member. How big is the sample?
- A2. 1,000.
- Q3. What is the difference between stratified and cluster sampling?
- A3. Stratified takes a random part from every group. Cluster takes all of some randomly chosen groups.

Where this is used in machine learning: Datasets are samples. When a dataset has very few examples of one class, such as fraud cases among normal payments, stratified sampling keeps the same proportion of each class in the training and test sets.

Lesson 2.5 Measures of Central Tendency: Mean, Median and Mode

`file: content/learn/math/02-statistics/05-central-tendency.md` `minutes: 12` `prerequisites: 2.2`

Goal: Calculate the mean, median and mode, and use them to fill in missing values.

The idea in plain words A **measure of central tendency** is a value that represents the centre point, or typical value, of a dataset. It summarises the data with one number. There are three: the mean, the median and the mode.

Mean (arithmetic mean). The sum of the values divided by the number of values.

```
\mu = \frac{\sum x}{N}
```

Example. Five heights in cm: 160, 172, 165, 168, 174.

μ = (160 + 172 + 165 + 168 + 174) / 5 = 839 / 5 = 167.8

Median. The middle value of the list. To find it, first sort the numbers from smallest to largest.

- Odd number of values: sorted list 160, 165, 168, 172, 174. The middle value is **168**.
- Even number of values: add a sixth height, 176, so the sorted list is 160, 165, 168, 172, 174, 176. There are two middle values, 168 and 172, so the median is their average: (168 + 172) / 2 = **170**.

Mode. The value that occurs most often. In 2, 3, 3, 5, 7 the mode is 3. If no number in the list is repeated, there is no mode.

Using central tendency to fill in missing values. Data pre-processing means preparing the data before feeding it into a machine learning model. Measures of central tendency are very useful for handling missing values in a dataset.

- **Mean:** replace a missing value with the mean if the data is uniformly distributed, meaning almost all the data points are spread in a similar way (a symmetric, bell-shaped spread).
- **Median:** replace a missing value with the median if the data is skewed (lopsided).
- **Mode:** replace a missing value with the mode for categorical values, and also when the data is skewed.

```
import pandas as pd
heights = pd.Series([160, 172, None, 168, 174])
heights = heights.fillna(heights.mean())   # fills the gap with the mean,
168.5
print(heights.mean(), heights.median())
```

Note In this small code example the known values are 160, 172, 168 and 174, so their mean is 168.5. That is the number used to fill the gap.

Key rules

- Mean = sum divided by count. Median = middle value of the sorted list. Mode = most frequent value.
- Symmetric data: use the mean. Skewed data: use the median. Categorical data: use the mode.

Practice

- Q1. Find the mean of 2, 4, 6, 8.
- A1. 20 / 4 = 5.

- Q2. Find the median of 3, 9, 1, 7, 5.
- A2. Sorted: 1, 3, 5, 7, 9. The median is 5.
- Q3. Which measure would you use to fill a missing value in a very skewed income column?
- A3. The median.

Where this is used in machine learning: Real datasets almost always have gaps. Filling them with the right measure of central tendency keeps the model from being misled.

Lesson 2.6 Measures of Variability: Range, Variance and Standard Deviation

`file: content/learn/math/02-statistics/06-variability.md` `minutes: 15` `prerequisites: 2.5`

Goal: Calculate the range, variance and standard deviation, and explain why the mean alone can mislead.

The idea in plain words Measures of variability tell you how the data is spread out in a dataset.

- **Range:** the difference between the largest and the smallest value. It gives a rough idea of the spread.
- **Variance:** a measure of how far each number in the set is from the mean, and therefore from every other number in the set.
- **Standard deviation:** the square root of the variance. It shows how spread out a group of numbers is from the mean.

```
\text{Range} = \text{Max value} - \text{Min value} \qquad \sigma^2 =
\frac{\sum (x - \mu)^2}{N} \qquad \sigma = \sqrt{\sigma^2}
```

Worked example. Two datasets:

- A: -5, 0, 5, 10, 15
- B: 3, 4, 5, 6, 7

Step 1: Mean. A: (-5 + 0 + 5 + 10 + 15) / 5 = 25 / 5 = 5. B: (3 + 4 + 5 + 6 + 7) / 5 = 25 / 5 = 5. Both means are 5.

Step 2: Range. A: 15 - (-5) = 20. B: 7 - 3 = 4.

Step 3: Variance.

- A: $[(-5-5)^2 + (0-5)^2 + (5-5)^2 + (10-5)^2 + (15-5)^2] / 5 = (100 + 25 + 0 + 25 + 100) / 5 = 250 / 5 =$ **50**
- B: $[(3-5)^2 + (4-5)^2 + (5-5)^2 + (6-5)^2 + (7-5)^2] / 5 = (4 + 1 + 0 + 1 + 4) / 5 = 10 / 5 =$ **2**

Step 4: Standard deviation. A: $\sqrt{50} \approx 7.07$. B: $\sqrt{2} \approx 1.41$.

Both datasets have the same mean, 5, but A is far more spread out than B. The mean alone hides this.


```
{"type":"plot","xDomain":[-8,18],"yDomain":[0,3],"points":[{"at":
[-5,2],"label":"A"},{"at":[0,2],"label":""},{"at":[5,2],"label":""},{"at":
[10,2],"label":""},{"at":[15,2],"label":""},{"at":[3,1],"label":"B"},{"at":
[4,1],"label":""},{"at":[5,1],"label":""},{"at":[6,1],"label":""},{"at":
[7,1],"label":""}],"vlines":[{"x":5,"label":"mean = 5"}],"checkpoints":
[[-5,2],[15,2],[3,1],[7,1]],"caption":"Dataset A (top row) is spread from -5
to 15. Dataset B (bottom row) is bunched between 3 and 7. Both have a mean of
5."}
```


Note These formulas divide by N, the number of values, as in the notes. When data is a sample of a bigger population, many books divide by N - 1 instead. The course returns to this when models are built.

```
import numpy as np
A = np.array([-5, 0, 5, 10, 15])
B = np.array([3, 4, 5, 6, 7])
print(A.max() - A.min(), np.var(A), np.std(A))   # 20 50.0 7.07...
print(B.max() - B.min(), np.var(B), np.std(B))   # 4 2.0 1.41...
```

Key rules

- Range = largest - smallest.
- Variance = the average of the squared distances from the mean.
- Standard deviation = $\sqrt{}$variance, in the same units as the data.
- A small standard deviation means the data is bunched near the mean. A large one means it is spread out.

Practice

- Q1. Find the range of 4, 9, 15.
- A1. 15 - 4 = 11.
- Q2. Find the variance of 2, 4, 6.
- A2. The mean is 4. (4 + 0 + 4) / 3 = 8 / 3 ≈ 2.67.

- Q3. A dataset has a variance of 25. What is its standard deviation?
- A3. √25 = 5.

Where this is used in machine learning: Spread tells you whether a feature carries useful variety. It is also used to rescale features so that all of them have a similar size before a model learns from them.

Lesson 2.7 Percentiles and Quantiles

`file: content/learn/math/02-statistics/07-percentiles-quantiles.md` `minutes: 12` `prerequisites: 2.5`

Goal: Read a percentile, find the quartiles of a dataset, and say how a percentile differs from a percentage.

The idea in plain words These measures show how the data points are distributed in a dataset, for example age, annual income or height.

Percentile. A percentile is a value on a scale of 100 that shows the percent of the data that is equal to or below it. If 190 cm is the 80th percentile in a group of 15 people, then 80 percent of the heights are at or below 190 cm. Note that a percentile is different from a percentage. A percentage is a part out of 100, like scoring 80 percent in a test. A percentile is a position within the data.

Quantile. A quantile is a measure that tells how many values in a dataset are above or below a certain limit. It divides the dataset into equally sized subgroups. The most common quantiles are the **quartiles**:

- Q1, the 25th percentile.
- Q2, the 50th percentile, which is the **median**.
- Q3, the 75th percentile.

Worked example. The heights, in cm, of 15 people, sorted:

150, 152, 155, 158, 160, 162, 165, 168, 170, 172, 175, 190, 192, 195, 198

- Percentile of 190: 12 of the 15 values are at or below 190, and 12 / 15 = 80 percent. So 190 is the **80th percentile**.
- Median (Q2): the middle, 8th value, which is **168**.
- Q1: the first value with at least 25 percent of the data at or below it. The 4th value, **158**, has 4 / 15 ≈ 27 percent.
- Q3: the first value with at least 75 percent at or below it. The 12th value, **190**, has 80 percent. The 11th value, 175, has only 73 percent.

```
{"type":"plot","xDomain":[145,205],"yDomain":[-1,2],"points":[{"at":
[150,0],"label":""},{"at":[152,0],"label":""},{"at":[155,0],"label":""},
{"at":[158,0],"label":""},{"at":[160,0],"label":""},{"at":
[162,0],"label":""},{"at":[165,0],"label":""},{"at":[168,0],"label":""},
{"at":[170,0],"label":""},{"at":[172,0],"label":""},{"at":
[175,0],"label":""},{"at":[190,0],"label":""},{"at":[192,0],"label":""},
{"at":[195,0],"label":""},{"at":[198,0],"label":""}],"vlines":
[{"x":158,"label":"Q1 = 158"},{"x":168,"label":"median = 168"},
{"x":190,"label":"Q3 = 190"}],"checkpoints":[[150,0],[168,0],[190,0],
[198,0]],"caption":"Fifteen heights on a number line with the quartiles
marked: Q1 at 158, the median at 168 and Q3 at 190."}
```


```
import numpy as np
heights = [150,152,155,158,160,162,165,168,170,172,175,190,192,195,198]
print(np.percentile(heights, [25, 50, 75]))   # [159.  168.  182.5]
```

Note Python gives 159 and 182.5 for Q1 and Q3 because it blends the two neighbouring values. Different tools use slightly different rules for quartiles, so small differences are normal.

Key rules

- The kth percentile is the value with k percent of the data at or below it.
- Quartiles split the data into four equal parts: Q1 is the 25th percentile, Q2 the median, Q3 the 75th.
- Percentile is a position in the data. Percentage is a part out of 100.

Practice

- Q1. In a set of 80 values, 60 are at or below x. What percentile is x?
- A1. 60 / 80 = 75 percent, so the 75th percentile.
- Q2. Which percentile is the median?
- A2. The 50th.
- Q3. Into how many equal parts do quartiles divide a dataset?
- A3. Four.

Where this is used in machine learning: Percentiles and quartiles help spot unusually large or small values (outliers) in a column before a model is trained.

Lesson 2.8 Correlation and Causation

`file: content/learn/math/02-statistics/08-correlation-causation.md` `minutes: 12` `prerequisites: 2.3`

Goal: Read a scatter plot to tell positive and negative correlation, and explain why correlation does not prove causation.

The idea in plain words Correlation is a measure of how strongly two variables are related in a dataset. It describes the relationship between features. But it does not mean that one event is the cause of the other.

Types of correlation

- **Positive correlation:** the two variables move together, so one rises when the other rises (directly proportional). Example: house price and number of rooms.
- **Negative correlation:** the variables move in opposite directions, so one falls when the other rises (inversely proportional). Example: house price and crime rate.


```
{"type":"scatter","xDomain":[0,6],"yDomain":[50,400],"xLabel":"Number of rooms","yLabel":"House price (thousands)","points":[[1,100],[2,150],[2,170],[3,200],[3,230],[4,260],[4,280],[5,330]],"line":{"slope":56.67,"intercept":45},"checkpoints":[[1,100],[3,200],[5,330]],"caption":"Positive correlation. Houses with more rooms tend to cost more, so the points rise from left to right."}
```



```
{"type":"scatter","xDomain":[0,8],"yDomain":[100,400],"xLabel":"Crime rate","yLabel":"House price (thousands)","points":[[1,320],[2,300],[3,270],[4,240],[5,200],[6,190],[7,150]],"line":{"slope":-28.57,"intercept":352.86},"checkpoints":[[1,320],[4,240],[7,150]],"caption":"Negative correlation. Where the crime rate is higher, house prices tend to be lower, so the points fall from left to right."}
```


Causation. In statistics, causation means that one event causes the other. It is a cause-and-effect relationship between two variables in a dataset.

Example: ice cream and temperature. How are ice cream sales affected by temperature? Sales rise as the temperature rises, and here the weather really does drive the sales.

```
{"type":"scatter","xDomain":[18,34],"yDomain":[0,110],"xLabel":"Temperature
(degrees Celsius)","yLabel":"Ice cream sales (units)","points":[[20,30],
[22,40],[24,45],[26,60],[28,70],[30,85],[32,95]],"line":
{"slope":5.536,"intercept":-83.2},"checkpoints":[[20,30],[26,60],
[32,95]],"caption":"Ice cream sales rise with temperature, a positive
correlation."}
```


Why correlation is not causation. Ice cream sales and sunburn cases both rise in summer, so they are correlated. But ice cream does not cause sunburn. Hot weather causes both. Always ask whether a third factor could explain the link.

Key rules

- Correlation measures how two variables move together. Its value, called r, runs from -1 (perfect negative) through 0 (none) to +1 (perfect positive).
- Correlation does not prove causation.

Practice

- Q1. As study hours go up, exam scores go up. Is this positive or negative correlation?
- A1. Positive.
- Q2. As the price of a product goes up, the number sold goes down. Which type is this?
- A2. Negative correlation.
- Q3. Ice cream sales and drowning cases both rise in summer. Does ice cream cause drowning?
- A3. No. Hot weather increases both. They are correlated, not causal.

Where this is used in machine learning: A correlation heatmap shows which features move with the target, such as house price. Strongly related features make good model inputs, but a link in the data does not prove a cause.

Lesson 2.9 Hypothesis Testing

`file: content/learn/math/02-statistics/09-hypothesis-testing.md` `minutes: 15` `prerequisites: 2.6`

Goal: State a null and an alternative hypothesis, and explain how data is used to decide between them.

The idea in plain words A **hypothesis** is an assumption made based on the observations of an experiment. **Hypothesis testing** is a method used to test the assumptions made in the experiment.

The two hypotheses

- **Null hypothesis (H0):** the commonly accepted fact.
- **Alternative hypothesis (Ha):** the opposite of the null hypothesis. It challenges the null hypothesis.

A historical example. Ptolemy proposed that the sun and the other planets revolve around the earth. That was the commonly accepted fact, so it plays the role of H0. Aryabhata proposed that the earth and the other planets revolve around the sun. That challenges it, so it plays the role of Ha.

A medical example. A pharmaceutical company has two drugs, Drug A and Drug B, for a headache. Group 1 takes Drug A and group 2 takes Drug B. The time taken for relief, in minutes:

- Drug A: 12, 8, 13, 10, 7. The average is 50 / 5 = 10 minutes.
- Drug B: 15, 12, 18, 16, 14. The average is 75 / 5 = 15 minutes.

Drug A takes 10 minutes on average and Drug B takes 15, so Drug A looks quicker. In the notes, the claim that Drug A is quicker than Drug B is written as the null hypothesis, and the claim that Drug B is quicker is the alternative.


```
{"type":"bars","xLabel":"Drug","yLabel":"Average minutes to relief","labels":["Drug A","Drug B"],"series":[{"name":"Average time","values":[10,15],"color":"accent"}],"checkpoints":[["Drug A",10],["Drug B",15]],"caption":"Average time to headache relief: Drug A takes 10 minutes and Drug B takes 15 minutes."}
```


Possible outcomes of a test. There are only two:

- **Reject** the null hypothesis, or
- **Fail to reject** the null hypothesis.

Note Many textbooks write the null hypothesis as no difference (both drugs take the same time on average) and the alternative as there is a difference. The idea is the same in both versions. H0 is the starting claim, Ha challenges it, and the data decides. The next steps use that standard form.

How the decision is made. Compare the gap between the averages, here 5 minutes, with the natural variation in the data. A statistical test turns this into a **p-value**: the probability of seeing a gap this large if H0 were true. A common rule is to reject H0 when the p-value

is below 0.05. For these numbers, a two-sample t-test gives a p-value of about 0.01, which is below 0.05, so we reject the null hypothesis of no difference. The data supports that the drugs differ, and Drug A is faster.

```
from scipy import stats
a = [12, 8, 13, 10, 7]
b = [15, 12, 18, 16, 14]
t, p = stats.ttest_ind(a, b)
print(p)    # about 0.01
```

Key rules

- H0 is the claim we start with. Ha challenges it.
- A test ends in one of two ways: reject H0, or fail to reject H0.
- A small p-value (below 0.05 is common) is evidence against H0.

Practice

- Q1. What are the two possible outcomes of a hypothesis test?
- A1. Reject the null hypothesis, or fail to reject the null hypothesis.
- Q2. Which hypothesis is the commonly accepted fact?
- A2. The null hypothesis.
- Q3. A test gives a p-value of 0.30. With the 0.05 rule, do we reject H0?
- A3. No. 0.30 is larger than 0.05, so we fail to reject H0.

Where this is used in machine learning: Hypothesis testing is used to decide whether a new model is really better than an old one, or whether a difference in results could be just chance.

18.9 Topic 3: Probability

Probability measures how likely things are. This topic follows Ezra's notes in order: what probability is, random variables, probability distributions, the normal distribution with skewness, and the Poisson distribution. One short lesson on probability rules and Bayes' theorem is added at the end.

Lesson 3.1 What Is Probability and Why Machine Learning Needs It

`file: content/learn/math/03-probability/01-what-is-probability.md` `minutes: 12` `prerequisites: 0.1`

Goal: Calculate the probability of simple events, place it on the 0 to 1 scale, and see two ways machine learning uses it.

The idea in plain words Probability is a branch of mathematics that deals with calculating the likelihood of a given event to occur. Simple examples: rolling a die, tossing a coin, and picking from a bag that contains balls of different colours.

```
P(\text{event}) = \frac{\text{number of ways the event can occur}}
{\text{total number of outcomes}}
```

Coin. Heads or tails, so 2 outcomes. P(Head) = 1 / 2 = 0.5.

Die. The possible outcomes are 1, 2, 3, 4, 5, 6. P(5) = 1 / 6 ≈ 0.17. For an even number there are three ways (2, 4, 6) out of 6, so P(even) = 3 / 6 = 1 / 2 = 0.5.

The probability scale. A probability value always lies between 0 and 1.

Value	Meaning
0	Impossible (0 percent)
0.5, or 1/2	Half chance (50 percent)
1	Certain (100 percent)

Examples with a die. What is the probability of getting a number less than 10? Every face is below 10, so the answer is **1**. What is the probability of getting a number less than 1? No face is below 1, so the answer is **0**.

Why probability matters in machine learning. Two main aspects:

1. The model's output. Some models, such as logistic regression, turn any number into a probability between 0 and 1 using an S-shaped curve.

```
{"type":"plot","xDomain":[-6,6],"yDomain":[-0.1,1.1],"functions":
[{"fn":"1/(1+exp(-x))","label":"S-shaped (logistic)
curve","color":"accent"}],"checkpoints":[[0,0.5],[2,0.8808],[-2,0.1192],
[4,0.982]],"caption":"The logistic curve. Whatever number goes in, the output
stays between 0 and 1, so it can be read as a probability. At 0 the output is
0.5."}
```

2. The loss function. A model is judged by how far its predictions are from the truth. For house prices, y is the true price and ŷ (read y-hat) is the predicted price. The loss averages the squared distances over n houses:

```
\text{Loss} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2
```

The smaller the loss, the closer the predictions are to the true values. Lesson 4.4 shows how a model reduces it.

What comes next in the course. The notes list further probability topics: maximum likelihood, Bayes' theorem, information theory, and cross entropy with information gain. These are taught inside the ML course while models are built from scratch.

Key rules

- P(event) = ways the event can happen / total outcomes.
- Probability is always between 0 and 1. 0 is impossible, 1 is certain.
- Machine learning uses probability in model outputs and in loss functions.

Practice

- Q1. What is the probability of rolling a 3 on a fair die?
- A1. 1 / 6 ≈ 0.17.
- Q2. What is the probability of rolling an odd number?
- A2. 3 / 6 = 0.5.
- Q3. What is the probability of an impossible event?
- A3. 0.

Where this is used in machine learning: A spam filter or a fraud detector does not just say yes or no. It first produces a probability, and then a cut-off turns it into a decision.

Lesson 3.2 Random Variables

`file: content/learn/math/03-probability/02-random-variables.md` `minutes: 10` `prerequisites: 3.1`

Goal: Explain what a random variable is, and tell discrete from continuous ones.

The idea in plain words A **random variable** is a numerical description of the outcomes of random events. In other words, a random variable maps the outcomes of random events to numerical values.

Example: tossing a coin. Call the random variable X. The random events are Head and Tail, and X turns them into numbers:

Random event	Value of X
Head	0
Tail	1

Written compactly, X = 0 if the toss is a head, and X = 1 if it is a tail.

Example: weight. Let Y be the weight of a random person in a class. We can then ask a probability question in numbers: what is the probability that the weight of a random person in the class is less than 60 kg? This is written P(Y < 60).

Application. An insurance company can use a random variable for the chance of a claim over a given time period.

Types of random variables

- **Discrete:** takes only distinct values. Example: the number shown when a die is rolled, or the number of heads in some coin tosses.
- **Continuous:** takes any value in a given range. Example: the weight of a random person in a class.

Key rules

- A random variable gives a number to every outcome of a random event.
- Discrete: separate values. Continuous: any value in a range.

Practice

- Q1. For a coin toss with X = 0 for Head and X = 1 for Tail, what is the value of X when the toss is a tail?
- A1. 1.
- Q2. Is the weight of a person a discrete or a continuous random variable?
- A2. Continuous.
- Q3. How do you write the probability that a person weighs less than 60 kg?
- A3. P(Y < 60).

Where this is used in machine learning: Every feature and every target in a dataset, such as house price, can be treated as a random variable.

Lesson 3.3 Probability Distribution of a Random Variable

`file: content/learn/math/03-probability/03-probability-distributions.md` `minutes: 12` `prerequisites: 3.2`

Goal: Build the probability distribution of a simple random variable and check that it is valid.

The idea in plain words The probability distribution of a random variable describes how the probabilities are distributed over the values of the random variable.

Example: tossing 3 coins. Let X be the number of heads when three coins are tossed. List all 8 equally likely outcomes:

Outcome	Heads	X
HHH	3	3
HHT	2	2
HTH	2	2
THH	2	2
HTT	1	1
THT	1	1
TTH	1	1
TTT	0	0

Now count how many outcomes give each value, and divide by 8. This is the distribution, shown as a table:

X (number of heads)	P(X = x) as a fraction	P(X = x) as a decimal
0	1/8	0.125
1	3/8	0.375
2	3/8	0.375
3	1/8	0.125

{"type":"bars","xLabel":"X = number of heads","yLabel":"P(X = x)","labels":["0","1","2","3"],"series":[{"name":"Probability","values":[0.125,0.375,0.375,0.125],"color":"accent"}],"checkpoints":[["0",0.125],["1",0.375],["2",0.375],["3",0.125]],"caption":"Probability distribution for the number of heads in three coin tosses. Two heads and one head are the most likely, each with probability 0.375."}

Check the distribution. The probabilities add up to 0.125 + 0.375 + 0.375 + 0.125 = 1. They must always add up to 1, because one of the outcomes is certain to happen.

Key rules

- A probability distribution lists every value of the random variable with its probability.
- Every probability is between 0 and 1, and all of them add up to exactly 1.

Practice

- Q1. In the 3-coin example, what is P(X = 0)?
- A1. 1 / 8 = 0.125.
- Q2. What is P(X is at least 2)?
- A2. 0.375 + 0.125 = 0.5.
- Q3. Two coins are tossed and X is the number of heads. Give the distribution.
- A3. X = 0 has 0.25, X = 1 has 0.5, X = 2 has 0.25.

Where this is used in machine learning: A classifier's output is a probability distribution over the possible classes, such as spam and not spam.

Lesson 3.4 The Normal Distribution and Skewness

`file: content/learn/math/03-probability/04-normal-distribution-skewness.md` `minutes: 15` `prerequisites: 3.3, 2.5`

Goal: Describe the normal distribution, use the 68-95-99.7 rule, and recognise positive and negative skew.

The idea in plain words A **normal distribution** is an arrangement of a dataset in which most of the data points lie in the middle range, and the rest taper off symmetrically toward either extreme. It is also known as the **Gaussian distribution**, or the **bell-shaped curve**. Heights of people are a classic example.

{"type":"plot","xDomain":[-4,4],"yDomain":[0,0.45],"functions":[{"fn":"exp(-x^2/2)/2.5066","label":"normal curve","color":"accent"}],"areas":[{"fn":"exp(-x^2/2)/2.5066","from":-1,"to":1}],"vlines":[{"x":-1,"label":"μ - σ"},{"x":0,"label":"μ"},{"x":1,"label":"μ + σ"}],"checkpoints":[[0,0.3989],[1,0.242],[2,0.054]],"caption":"The normal curve is symmetric around the mean μ. The shaded area between μ - σ and μ + σ holds about 68 percent of the data."}

The 68-95-99.7 rule. Measure distance from the mean μ in steps of the standard deviation σ:

Range	Share of the data
μ - σ to μ + σ	about 68% (34.1% on each side of the mean)
μ - 2σ to μ + 2σ	about 95% (another 13.6% on each side)
μ - 3σ to μ + 3σ	about 99.7% (another 2.1% on each side)

Worked example. Suppose adult heights have a mean of 160 cm and a standard deviation of 10 cm. About 68 percent of people are between 150 and 170 cm. About 95 percent are between 140 and 180 cm. About 99.7 percent are between 130 and 190 cm.

Skewness. Data is considered **skewed** when its distribution curve looks distorted, pulled to the left or to the right. Here are the three shapes:

{"type":"plot","xDomain":[0,8],"yDomain":[0,0.45],"functions":[{"fn":"(8-x)*exp(-(8-x))","label":"negatively skewed","color":"accent2"}],"vlines":[{"x":6,"label":"mean"},{"x":6.32,"label":"median"},{"x":7,"label":"mode"}],"checkpoints":[[7,0.3679]],"caption":"Negatively skewed data. The long tail is on the left, and the order from left to right is mean, median, mode."}

{"type":"plot","xDomain":[0,8],"yDomain":[0,0.45],"functions":[{"fn":"x*exp(-x)","label":"positively skewed","color":"accent2"}],"vlines":[{"x":1,"label":"mode"},{"x":1.68,"label":"median"},{"x":2,"label":"mean"}],"checkpoints":[[1,0.3679]],"caption":"Positively skewed data. The long tail is on the right, and the order from left to right is mode, median, mean."}

Shape	Order of the three averages
Negatively skewed (tail on the left)	mean < median < mode
Normal (no skew)	mean = median = mode
Positively skewed (tail on the right)	mode < median < mean

Example. Average income of people in different cities is positively skewed: most people earn moderate incomes, but a few very high earners stretch the tail to the right and pull the mean upwards. That is why the median is the better average to use for skewed data, as Lesson 2.5 said.

Key rules

- Normal distribution: bell-shaped, symmetric, mean = median = mode.
- 68 percent of the data lies within 1σ of the mean, 95 percent within 2σ, and 99.7 percent within 3σ.
- Skew is named after the side the long tail is on.

Practice

- Q1. In a normal distribution, how do the mean, median and mode compare?
- A1. They are all equal.
- Q2. Incomes in a city are positively skewed. Which is larger, the mean or the median?
- A2. The mean.
- Q3. About what percent of a normal distribution lies within 2 standard deviations of the mean?
- A3. About 95 percent.

Where this is used in machine learning: Many models assume that errors or features are roughly normal. Checking for skew tells you whether a feature needs to be transformed, and which average to use for missing values.

Lesson 3.5 The Poisson Distribution

`file: content/learn/math/03-probability/05-poisson-distribution.md` `minutes: 15` `prerequisites: 3.3`

Goal: Use the Poisson formula to find the probability of a number of events in a fixed interval.

The idea in plain words The **Poisson distribution** is a probability distribution that measures how many times an event is likely to occur within a specified period of time. It is used for independent events that occur at a constant rate within a given interval of time.

Examples

- The number of accidents occurring in a city from 6 pm to 10 pm.
- The number of patients arriving in an emergency room between 10 pm and 12 pm.
- How many views your blog gets in a day.

The formula

```
P(x) = \frac{e^{-\lambda}\,\lambda^{x}}{x!}
```

- x: the number of times the event occurs.
- P(x): the probability of exactly x events.
- λ (lambda): the mean (average) number of events in the interval.
- x!: the factorial of x, for example 4! = 4 × 3 × 2 × 1 = 24, and 0! = 1.
- e: Euler's number, about 2.71828.

Worked example. An emergency room receives on average λ = 3 patients per hour. What is the probability that exactly 2 arrive in the next hour?

$P(2) = e^{-3} \times 3^2 / 2! = 0.0498 \times 9 / 2 = 0.224$

So there is about a 22.4 percent chance. The whole distribution for λ = 3:

x	0	1	2	3	4	5	6	7	8
P(x)	0.0498	0.1494	0.2240	0.2240	0.1680	0.1008	0.0504	0.0216	0.0081


```
{"type":"bars","xLabel":"x = patients arriving in one
hour","yLabel":"P(x)","labels":
["0","1","2","3","4","5","6","7","8"],"series":[{"name":"lambda =
3","values":
[0.0498,0.1494,0.224,0.224,0.168,0.1008,0.0504,0.0216,0.0081],"color":"accent
"}],"checkpoints":[["2",0.224],["3",0.224],["4",0.168]],"caption":"Poisson
distribution with an average of 3 events per interval. The most likely counts
are 2 and 3."}
```

How λ changes the shape. Compare λ = 1, 5 and 10 (for example the average number of accidents occurring in an interval). As λ grows, the peak moves to the right, becomes lower, and the curve spreads out and looks more like a bell.


```
{"type":"bars","style":"lines","xLabel":"x = number of events
occurring","yLabel":"P(x)","labels":
["0","1","2","3","4","5","6","7","8","9","10","11","12","13","14","15"],"seri
es":[{"name":"lambda = 1","values":
[0.3679,0.3679,0.1839,0.0613,0.0153,0.0031,0.0005,0.0001,0,0,0,0,0,0,0,0],"co
lor":"accent"},{"name":"lambda = 5","values":
[0.0067,0.0337,0.0842,0.1404,0.1755,0.1755,0.1462,0.1044,0.0653,0.0363,0.0181
,0.0082,0.0034,0.0013,0.0005,0.0002],"color":"accent2"},{"name":"lambda =
10","values":
[0,0.0005,0.0023,0.0076,0.0189,0.0378,0.0631,0.0901,0.1126,0.1251,0.1251,0.11
37,0.0948,0.0729,0.0521,0.0347],"color":"text"}],"checkpoints":
[["1",0.3679,"lambda = 1"],["5",0.1755,"lambda = 5"],["10",0.1251,"lambda =
10"]],"caption":"Poisson curves for averages of 1, 5 and 10 events. The peaks
are about 0.37 at x = 1, 0.18 at x = 5, and 0.13 at x = 10."}
```


Application in data science. Poisson counts are used to model the number of accidents occurring in a particular region or area, so that busy and quiet areas can be compared.

```
from scipy.stats import poisson
print(poisson.pmf(2, 3))   # 0.2240...
```

Key rules

- Use Poisson for counting independent events at a constant average rate in a fixed interval.
- P(x) = e^(-λ) × λ^x / x!, where λ is the average number of events.

Practice

- Q1. Cars pass a point at an average of 2 per minute. What is the probability that none pass in a minute?
- A1. $P(0) = e^{-2} \times 2^0 / 0! = e^{-2} \approx 0.135$.
- Q2. What does λ stand for?
- A2. The average number of events in the interval.
- Q3. Is the number of emails you receive per hour a good fit for the Poisson distribution?
- A3. Usually yes, because the emails are counted events arriving at a roughly constant rate.

Where this is used in machine learning: Count data, such as visits, accidents or arrivals, is modelled with the Poisson distribution, and it appears again in models that predict counts.

Lesson 3.6 Probability Rules and Bayes' Theorem

`file: content/learn/math/03-probability/06-rules-bayes.md` `minutes: 12` `prerequisites: 3.1`

Goal: Use the basic probability rules and Bayes' theorem on a medical test example.

The idea in plain words The complement rule. The chance that an event does not happen is 1 minus the chance that it does. P(not A) = 1 - P(A). On a die, P(not 6) = 1 - 1/6 = 5/6.

Independent events. If one event does not affect the other, multiply their probabilities. Two heads in a row: P = 1/2 × 1/2 = 1/4. Three heads in a row: 1/8 = 0.125, which matches the 3-coin table in Lesson 3.3.

Either of two events. If two events cannot both happen, add their probabilities. P(die shows 1 or 2) = 1/6 + 1/6 = 2/6.

Conditional probability. P(A given B) is the probability of A once we know B has happened:

```
P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}
```

Bayes' theorem turns a conditional probability around:

```
P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}
```

Worked example. Does a person have diabetes or not? Suppose 1 percent of people have diabetes. A test is positive for 90 percent of people who have it, and also positive for 5 percent of people who do not. A person tests positive. What is the chance they really have diabetes?

- P(positive) = 0.90 × 0.01 + 0.05 × 0.99 = 0.009 + 0.0495 = 0.0585
- P(diabetes given positive) = 0.009 / 0.0585 ≈ 0.154

The answer is only about 15.4 percent. The disease is rare, so most positive results come from the large healthy group. Probability helps you avoid the trap of reading a positive test as a certainty.

Key rules

- P(not A) = 1 - P(A).

- Independent events: multiply. Events that cannot both happen: add.
- Bayes: P(A given B) = P(B given A) × P(A) / P(B).

Practice

- Q1. What is the probability of not rolling a 6?
- A1. 5 / 6.
- Q2. What is the probability of rolling two sixes in a row?
- A2. 1/6 × 1/6 = 1/36.
- Q3. Two events are independent, with probabilities 0.5 and 0.2. What is the chance of both?
- A3. 0.5 × 0.2 = 0.1.

Where this is used in machine learning: Bayes' theorem is the basis of the Naive Bayes classifier, which is often used for spam filtering.

18.10 Topic 4: Calculus (only what machine learning needs)

Ezra's notes introduce calculus in one page: differential calculus splits the data into smaller pieces and looks at how it changes, integral calculus adds small pieces into a large one, and gradient descent comes from it. This topic was written by the author to turn that page into five short lessons. It keeps to what a beginner needs to understand how a model is trained.

Lesson 4.1 Functions, Graphs and Slope

`file: content/learn/math/04-calculus/01-functions-slope.md` `minutes: 10` `prerequisites: 0.1`

Goal: Read a function as a rule, calculate a slope, and see why a curve needs smaller and smaller steps.

The idea in plain words A **function** is a rule that takes an input x and gives an output, written f(x). For example $f(x) = x^2$ squares its input. A machine learning model is a function: data goes in and a prediction comes out.

The **slope** tells how fast the output changes when the input changes:

```
\text{slope} = \frac{\text{change in } y}{\text{change in } x} =
\frac{\text{rise}}{\text{run}}
```

For a straight line the slope is the same everywhere. The line through (1, 3) and (3, 7) has slope (7 - 3) / (3 - 1) = 2.

A curve is different, because its slope keeps changing. Take $f(x) = x^2$ and ask for the slope at x = 1. Measure it over smaller and smaller steps:

Step from x = 1 to	Slope = change in x^2 / change in x
2	(4 - 1) / 1 = 3
1.1	(1.21 - 1) / 0.1 = 2.1
1.01	(1.0201 - 1) / 0.01 = 2.01

The slopes approach 2 as the step shrinks. Splitting into tiny pieces and watching the change is exactly what **differential calculus** does. The slope at one single point is called the **derivative**, the subject of the next lesson.


```
{"type":"plot","xDomain":[-1,3],"yDomain":[-1,6],"functions":
[{"fn":"x^2","label":"f(x) = x²","color":"accent"},{"fn":"2*x-
1","label":"slope 2 at x = 1","color":"accent2"}],"points":[{"at":
[1,1],"label":"(1, 1)"},{"at":[2,4],"label":"(2, 4)"}],"checkpoints":[[1,1],
[2,4],[0,-1],[2,3]],"caption":"The curve y = x² and its tangent line y = 2x -
1 at the point (1, 1). The tangent has slope 2, the slope of the curve at
that point."}
```


Key rules

- Slope = change in y divided by change in x.
- A straight line has one slope. A curve has a different slope at every point.
- Shrinking the step gives the slope at a single point.

Practice

- Q1. Find the slope of the line through (0, 1) and (2, 5).
- A1. (5 - 1) / (2 - 0) = 2.
- Q2. What is the slope of the line y = 3x + 2?
- A2. 3.
- Q3. For $y = x^2$, find the slope between x = 2 and x = 3.
- A3. (9 - 4) / 1 = 5.

Where this is used in machine learning: A model is a function. The slope tells us how much the prediction changes when we nudge an input or a weight.

Lesson 4.2 The Derivative and Its Rules

`file: content/learn/math/04-calculus/02-derivative-rules.md` `minutes: 12` `prerequisites: 4.1`

Goal: Differentiate simple functions with the standard rules, and use the derivative to find a minimum.

The idea in plain words The **derivative** of f at x is the slope of the curve at that point. It is what the slope over a step h settles to as the step shrinks to zero:

```
f'(x) = \lim_{h \to 0}\frac{f(x+h) - f(x)}{h}
```

It is also written df/dx. Check it for f(x) = x²: (x + h)² - x² = 2xh + h², and dividing by h gives 2x + h. As h shrinks to 0 this becomes 2x. So f′(x) = 2x, and at x = 1 the slope is 2, just as the last lesson found.

The rules you need

Rule	Statement	Example
Constant	the derivative of a constant is 0	derivative of 7 is 0
Power	d/dx of x^n is n × x^(n-1)	d/dx of x^3 is $3x^2$
Constant multiple	d/dx of c × f is c × f′	d/dx of $5x^2$ is 10x
Sum	(f + g)′ = f′ + g′	d/dx of (x^2 + x) is 2x + 1
Chain	d/dx of f(g(x)) is f′(g(x)) × g′(x)	d/dx of $(2x + 1)^2$ is 2(2x + 1) × 2 = 8x + 4

Worked examples

- f(x) = $3x^2$ + 2x + 5 gives f′(x) = 6x + 2.
- f(x) = $(2x + 1)^2$. By the chain rule, f′(x) = 2(2x + 1) × 2 = 8x + 4. Check by expanding: $(2x + 1)^2$ = $4x^2$ + 4x + 1, and its derivative is 8x + 4.

Finding a minimum. At the lowest point of a smooth curve the slope is 0. Take f(x) = (x - 3)². Then f′(x) = 2(x - 3). Setting it to 0 gives x = 3, where f(3) = 0, the minimum. To the left of 3 the slope is negative (the curve falls), for example f′(1) = -4. To the right it is positive (the curve rises), for example f′(5) = 4.


```
{"type":"plot","xDomain":[0,6],"yDomain":[-1,10],"functions":[{"fn":"(x-3)^2","label":"f(x) = (x - 3)²","color":"accent"}],"points":[{"at":[3,0],"label":"minimum: slope 0"},{"at":[1,4],"label":"slope -4"},{"at":[5,4],"label":"slope +4"}],"checkpoints":[[3,0],[1,4],[5,4]],"caption":"The curve y = (x - 3)². The slope is negative on the left of the minimum, zero at x = 3, and positive on the right."}
```


Key rules

- The derivative is the slope of the curve at a point.
- Power rule: x^n becomes n × x^(n-1). Constants disappear. Sums are differentiated term by term.
- Where the derivative is 0, the curve is flat: that is where a minimum or maximum can be.

Practice

- Q1. Find the derivative of x³.
- A1. 3x².
- Q2. Find the derivative of 5x² - x.
- A2. 10x - 1.
- Q3. At which x does (x - 2)² reach its minimum?
- A3. At x = 2, because the derivative 2(x - 2) is 0 there.

Where this is used in machine learning: Training searches for the lowest point of the loss. The derivative tells which way is downhill and where the bottom is.

Lesson 4.3 Partial Derivatives and the Gradient

`file: content/learn/math/04-calculus/03-partial-derivatives-gradient.md` `minutes: 10` `prerequisites: 4.2, 1.2`

Goal: Take a partial derivative, build a gradient vector, and say which way it points.

The idea in plain words A real model has many numbers to adjust, not one. A function of several variables, such as f(x, y), has one slope for each variable. A **partial derivative** differentiates with respect to one variable and treats the others as constants.

For f(x, y) = x² + 3y:

- The partial derivative with respect to x is 2x. (The 3y part is treated as a constant, so it disappears.)
- The partial derivative with respect to y is 3.

The **gradient** collects all the partial derivatives into one vector, written ∇f:

```
\nabla f = \left(\frac{\partial f}{\partial x},\ \frac{\partial f}{\partial y}\right)
```

At the point (2, 1), the gradient of x² + 3y is (4, 3).

The key fact. The gradient points in the direction in which the function **increases fastest**. Walking in the opposite direction, along **minus the gradient**, goes downhill fastest.

Worked example. For f(x, y) = x² + y², the gradient is (2x, 2y). At the point (3, 4) it is (6, 8), which has magnitude 10 and points away from the lowest point, the origin. Minus the gradient, (-6, -8), points back toward the origin.


```
{"type":"plot","xDomain":[-5,7],"yDomain":[-5,9],"vectors":[{"from":[3,4],"to":[6,8],"label":"gradient (6, 8): uphill","color":"accent"},{"from":[3,4],"to":[-3,-4],"label":"minus gradient (-6, -8): downhill","color":"accent2"}],"points":[{"at":[3,4],"label":"(3, 4)"},{"at":[0,0],"label":"lowest point (0, 0)"}],"checkpoints":[[6,8],[-3,-4],[0,0]],"caption":"For f(x, y) = x² + y² at the point (3, 4), the gradient (6, 8) points uphill and its opposite points downhill, through the lowest point at the origin."}
```


Key rules

- A partial derivative treats every other variable as a constant.
- The gradient is the vector of all partial derivatives.
- The gradient points uphill. Minus the gradient points downhill.

Practice

- Q1. Find the partial derivative of x²y with respect to x.
- A1. 2xy.
- Q2. Find the gradient of x² + y² at (1, 2).
- A2. (2, 4).
- Q3. To make f smaller, should we move along the gradient or against it?

- A3. Against it.

Where this is used in machine learning: The loss depends on many weights at once. The gradient tells us, for every weight, how to change it to reduce the loss.

Lesson 4.4 Gradient Descent From Scratch

`file: content/learn/math/04-calculus/04-gradient-descent.md` `minutes: 15` `prerequisites: 4.2, 3.1, 1.3`

Goal: Train a one-weight model by hand with gradient descent, and see what the learning rate does.

The idea in plain words Training means adjusting the model until its loss is small. Gradient descent does it with one repeated step: find the slope of the loss, then move the weight a little in the downhill direction.

```
w_{\text{new}} = w - \eta \cdot \frac{dL}{dw}
```

Here w is the weight, L is the loss, and η (eta) is the **learning rate**, the size of each step.

Worked example. A model predicts ŷ = w × x. The data is x = 1, 2, 3 and the true answers are y = 2, 4, 6, so the best weight is w = 2. Use the loss from Lesson 3.1, the average squared error:

```
L(w) = \frac{1}{3}\left[(2 - w)^2 + (4 - 2w)^2 + (6 - 3w)^2\right] =
\frac{14}{3}(2 - w)^2
```

By the power and chain rules, the slope of the loss is:

```
\frac{dL}{dw} = \frac{28}{3}(w - 2)
```

Start at w = 0 with a learning rate of η = 0.1:

Step	w	Loss L(w)	Slope dL/dw	Next w = w - 0.1 × slope
0	0	18.667	-18.667	1.867
1	1.867	0.083	-1.244	1.991
2	1.991	0.0004	-0.083	1.999

The weight races toward 2 and the loss toward 0. Each step shrinks the distance to the answer by the same factor, 1 - 0.1 × 28/3 ≈ 0.067.

```
{"type":"plot","xDomain":[-0.5,4.5],"yDomain":[-1,30],"functions":
[{"fn":"4.6667*(2-x)^2","label":"loss L(w)","color":"accent"}],"points":
[{"at":[0,18.667],"label":"start: w = 0"},{"at":[1.867,0.083],"label":"after
1 step"},{"at":[2,0],"label":"minimum: w = 2"}],"checkpoints":[[0,18.667],
[1.867,0.083],[2,0]],"caption":"The loss curve L(w). Gradient descent starts
at w = 0 on the steep left side and slides down to the minimum at w = 2 in a
few steps."}
```


```
import numpy as np
x = np.array([1, 2, 3])
y = np.array([2, 4, 6])
w, lr = 0.0, 0.1
for step in range(4):
    grad = np.mean(-2 * x * (y - w * x))   # derivative of the loss
    w = w - lr * grad
    print(step + 1, round(w, 4))
# prints 1.8667, 1.9911, 1.9994, 2.0
```

The learning rate matters

Learning rate	What happens
Very small, such as 0.001	Safe but very slow
Well chosen, such as 0.1 here	Fast and steady
Too large, such as 0.25 here	Each step overshoots by more than before, and the weight flies away from 2

For this problem, each step multiplies the distance to 2 by 1 - η × 28/3. At η = 0.25 that factor is -1.33, so the error grows every step instead of shrinking.

Key rules

- Gradient descent: new weight = weight - learning rate × slope of the loss.
- A negative slope makes the weight go up. A positive slope makes it go down.
- The learning rate must be small enough to avoid overshooting.

Practice

- Q1. Write the gradient descent update rule in words.
- A1. The new weight is the old weight minus the learning rate times the slope of the loss.

- Q2. The slope of the loss at the current weight is negative. Does the weight go up or down?
- A2. Up, because we subtract a negative number.
- Q3. What can happen if the learning rate is too large?
- A3. The steps overshoot the minimum and the loss can grow instead of shrinking.

Where this is used in machine learning: This is what training means. Almost every model, from linear regression to deep networks, is trained by repeating this step on many weights at once.

Lesson 4.5 Integrals and Area Under a Curve

`file: content/learn/math/04-calculus/05-integrals-area.md` `minutes: 10` `prerequisites: 4.2, 3.4`

Goal: Explain an integral as an area, and connect it to probability.

The idea in plain words Integral calculus adds many small pieces together into one large whole. The most common picture is the **area under a curve**: cut the area into thin strips, find each strip's area, and add them up. The thinner the strips, the better the answer.

Worked example. The area under y = x² from x = 0 to x = 2. With 4 strips of width 0.5, using the height at the right-hand end of each strip: heights 0.25, 1, 2.25, 4. The area is about 0.5 × (0.25 + 1 + 2.25 + 4) = 3.75. That overestimates. With more and more strips it settles at the exact value, written as an integral:

```
\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} \approx 2.67
```

The rule used is that the integral of x^n is x^(n+1) / (n + 1). Integrals and derivatives are opposites: differentiating x³ / 3 gives back x².

```
{"type":"plot","xDomain":[-0.5,3],"yDomain":[-0.5,6],"functions":
[{"fn":"x^2","label":"y = x²","color":"accent"}],"areas":
[{"fn":"x^2","from":0,"to":2}],"points":[{"at":[2,4],"label":"(2,
4)"}],"checkpoints":[[1,1],[2,4]],"caption":"The shaded area under y = x²
between x = 0 and x = 2 equals 8/3, about 2.67."}
```

Link to probability. For a continuous random variable, such as height, probability is an area under its curve. The total area under any probability curve is 1. The area under the normal curve between μ - σ and μ + σ is about 0.683, which is the 68 percent from Lesson 3.4. For a discrete variable, such as the Poisson counts, we add up bar heights instead of finding an area.

Key rules

- An integral adds many small pieces into a whole, often the area under a curve.
- The integral of x^n is x^(n+1) / (n + 1), plus a constant when no limits are given.
- Total area under a probability curve = 1.

Practice

- Q1. Find the integral of 2x from 0 to 3.
- A1. x^2 evaluated from 0 to 3 is 9.
- Q2. What is the integral of x^2 without limits?
- A2. $x^3 / 3 + C$.
- Q3. What is the total area under any probability curve?
- A3. 1.

Where this is used in machine learning: Integrals turn probability curves into probabilities. For example, the chance that a person's height is between two values is the area under the height curve between them.

18.11 Notes on the source, the author's additions, and what to confirm

The lessons above follow the order of Ezra's handwritten notes: the big picture, linear algebra (vectors, then matrices), statistics, then probability. Calculus comes last because the notes only introduce it.

Written by the author, not in the notes: Lesson 1.9, Lesson 3.6, and all of Topic 4. Smaller additions: the worked example for (-3, 2) in Lesson 1.2, the projection example, the outcome check in Lesson 2.9 (p-value of about 0.01), the quartile example in Lesson 2.7, the 160 cm and 10 cm example in Lesson 3.4, and the $\lambda = 3$ example in Lesson 3.5.

Handwriting that could not be read with certainty, and what was done. Ezra should check each against the notebook.

- Vector addition, subtraction and scaling (Lesson 1.3): the numbers were hard to read, so clean examples were used. The example -0.5 × (3, 3) was kept as written.
- Matrix addition and multiplication (Lesson 1.8): the written results were partly unclear, so every result was recomputed. The subtraction example matches the notes.
- Cross product (Lesson 1.5): the answer in the notes looks like -4k, which matches (2, 3, 0) × (4, 4, 0).

- Mean of heights (Lesson 2.5): the notes show 167.8, which fits 160, 172, 165, 168, 174. The sixth height in the even-length median example (176) was not readable, so it was chosen.
- Dataset B in Lesson 2.6: the notes give a range of 4 and a variance of 2, which fit 3, 4, 5, 6, 7.
- Percentile example (Lesson 2.7): the 15 heights were chosen so that 190 is the 80th percentile, as in the notes.
- Pros and cons for stratified and cluster sampling (Lesson 2.4): partly unreadable, so the standard ones were written.
- Hypothesis example (Lesson 2.9): the notes state the null hypothesis as Drug A being quicker. The lesson keeps that and adds a note on the usual no-difference form.
- Die questions in Lesson 3.1: the second question is read as number less than 1, with answer 0.
- Normal curve percentages: the notes show roughly 68, 95 and 99.7 percent. The lesson uses these.
- Poisson probabilities for λ = 1, 5 and 10 (Lesson 3.5): the values written beside the curves were unclear, so exact values were used.

18.12 Lesson index and build checklist

Topic	Lessons	Total minutes
0 Introduction	1 (0.1)	6
1 Linear Algebra	9 (1.1 to 1.9)	93
2 Statistics	9 (2.1 to 2.9)	104
3 Probability	6 (3.1 to 3.6)	76
4 Calculus	5 (4.1 to 4.5)	57

There are 30 lessons, about 5.6 hours in total. The `/learn` overview shows these five topics in this order.

Checklist for the build agent

- Create one file per lesson at the path in its `file` field, with the body copied exactly from **Goal** to the last answer.

- Convert each `latex` block to a display formula between double dollar signs, and render with KaTeX.
- Convert every `A` bullet in a Practice list into a hidden answer with a **Show answer** button.
- Draw every `graph` block as specified in Section 18.5, then check each checkpoint and fix any graph that misses one.
- Link each lesson's prerequisites, and add previous and next links following the order of this section.
- Do not add, remove, shorten or reword any lesson text, number or formula.